Interference Sparsity-Aware Turbo Receiver for HF Skywave Massive MIMO

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***Abstract*—In this paper, we propose a low complexity turbo receiver for high frequency (HF) skywave massive multiple-input multiple-output (MIMO) systems. We first introduce the beam based channel model (BBCM) with uniform sampling for directional cosine. By leveraging the BBCM, we reveal the interference sparsity of HF skywave massive MIMO systems, which is defined as the asymptotic sparsity of the channel Gram matrix. Exploiting the interference sparsity, we provide a condition of extracting sufficient observation for signal detection. Motivated by this condition, we construct the interference user terminal (UT) set (IUS) and extract the observation vector from the received signal after matched filtering (MF) for each UT. Then, a low-dimensional interference sparsity-aware detector (ISD) is separately designed for each UT by minimizing the mean-squared error (MSE), and the interference sparsity-aware turbo receiver (ISTR) is subsequently formulated using ISDs. Under a relaxed version of the condition for sufficient observation selection, we prove the optimality of the ISTR. Further, we develop an efficient implementation of the ISTR, involving approximate computation of the ISD, the signal reconstructed by ISD and the channel Gram matrix. Moreover, an efficient construction of IUS using the statistical channel state information (CSI) is also proposed. Simulation results confirm that the proposed ISTR achieves excellent performance with relatively low complexity.**

***Index Terms*—Massive MIMO, HF skywave communications, interference sparsity, turbo receiver.**

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I. Introduction

A. Background

With the continuous increase in the demand for global coverage [1], high frequency (HF) skywave communications, operating within the frequency range of 3 to 30 MHz, have surfaced as a technology of a great promise. By virtue of ionosphere refraction, HF communications are capable of facilitating over-the-horizon communication [2]. Nevertheless, in traditional point-to-point HF communication scenarios, the data rates remain relatively low. This is attributed to the restricted spectrum resources and the intricate ionosphere conditions. In an attempt to address this issue, multiple-input multiple-output (MIMO) technology has been explored for its application in HF communications in several studies [3], [4], where certain performance enhancements have been achieved.

By deploying a large number of antennas at the base station (BS) to serve multiple user terminals (UTs), massive MIMO technology plays a significant role in the fifth Generation (5G) mobile communication systems and is expected to play its important role in the sixth Generation (6G) systems [5]. Consequently, introducing massive MIMO into HF skywave communications holds profound significance. The concept of HF skywave massive MIMO has been first proposed in [6], confirming the effectiveness of massive MIMO for HF skywave communications. Specifically, for HF skywave massive MIMO, downlink robust precoding [7], [8], uplink channel estimation [9], [10] and signal detection [11], [12] have been thoroughly studied.

B. Motivation and Challenges

For massive MIMO transmission, receiver design is a crucial issue. The optimal receiver, i.e., the maximum likelihood (ML) receiver [13] and some near-optimal nonlinear receivers [14], [15], suffer unacceptable complexity for massive MIMO and high-order modulation. In addition, nonlinear receivers within the framework of Bayesian inference have been widely investigated, e.g., belief propagation (BP) [16], expectation propagation (EP) [17], approximate message passing (AMP) [18] and information geometry approach (IGA) [19], while their performance and computational complexities highly depend on the convergence behaviors. As an alternative, linear receivers are also extensively applied in various communication scenarios, where the matched filtering (MF), zero-forcing (ZF), and minimum mean-squared error (MMSE) receivers are optimal for maximizing the

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received signal-to-noise ratio (SNR), signal-to-interference ratio (SIR), and signal-to-interference-plus-noise ratio (SINR), respectively [20]. To avoid high dimensional matrix inversions for ZF or MMSE receiver in massive MIMO systems, there are several approximate approaches [21], [22], [23]. However, the above linear receivers acheive low performance especially when with ill-conditioned channel matrix [24].

By exchanging soft information between a soft-input soft-output (SISO) detector and a SISO decoder, the turbo receiver can significantly improve the bit-error rate (BER) performance with moderate implementation complexity for practice [25]. The principle of the turbo receiver is similar to that of turbo codes [26]. It was first proposed in [27] using ML-based techniques. Moreover, turbo receivers can be naturally combined with popular detection techniques, such as MMSE [28], [29], EP [30], [31], AMP [32], [33], etc. Over the decades, turbo receivers have demonstrated significant value in numerous communication systems [34], [35], [36], [37], including massive MIMO communications [32], [38]. Nevertheless, the implementation of massive MIMO turbo receiver is limited by its high complexity, where both high-dimensional matrix operations and iterations are required.

C. Related Works

For massive MIMO communications, limited angle spread leads to remarkable sparsity of the beam domain channel. Some works utilize the beam domain channel sparsity for receiver designs [39], [40], [41], [42]. In [39], a low-dimensional ZF detector followed by a nonlinear post-processing scheme is developed with channel matrix compression using dominant beam selection, which provides a tradeoff between performance, complexity, and required memory. Besides, [40] propsoes a computationally efficient architecture, where a local MMSE detector is designed with the selection of a contiguous set of beams for each UT. To alleviate the performance degradation caused by information loss in the local MMSE detector, [41] proposes to combine the information from overlapped local MMSE filters. Moreover, a layered BP based detector is proposed in [42], where a concatenated structure of beam domain and space domain BP layers is utilized.

In the context of HF skywave massive MIMO, the beam domain channel sparsity also has been introduced for low-complexity receiver designs [11], [12]. In [11], separate and joint versions of Slepian transform based detectors are developed, where the modulated Slepian sequences are designed by employing the index set for non-zero beams of each UT. Moreover, [12] develops space and space-frequency domain beam structured detectors with low-dimensional beam domain detectors, and its asymptotic optimality is proved. As a generalization of [12], a beam structured turbo receiver (BSTR) with windowing is proposed in [43]. The BSTR incorporates the beam structured detector into the turbo receiver, achieving a better performance-complexity tradeoff. However, these works only focus on utilizing the beam domain channel sparsity for receiver designs, while other potential features of HF skywave massive MIMO systems are not fully exploited.

D. Contributions

In addition to the beam domain channel sparsity, the space domain channel between UTs without overlapping in the beam domain tends to be orthogonal as the number of antennas at the BS approaches infinity [6], resulting in the asymptotic sparsity of the channel Gram matrix, denoted as *interference sparsity*. Interestingly, this property provides a new perspective for low-complexity receiver design. In this paper, we investigate the turbo receiver design for HF skywave massive MIMO exploiting interference sparsity. We propose the interference sparsity-aware turbo receiver (ISTR) with reliable performance and low complexity. The main contributions are summarized as follows.

- Leveraging the beam based channel model (BBCM) and the sparsity of beam domain channels, we reveal the asymptotic sparsity of the channel Gram matrix for HF skywave massive MIMO systems. Since the magnitude of non-diagonal elements in the channel Gram matrix can serve as the interference level among UTs, we define the sparsity of the channel Gram matrix as the interference sparsity, which is the key feature for the design of the proposed ISTR.
- For signal detection using the a priori information in turbo receivers, we propose a condition for sufficient observation selection using the interference sparsity. Motivated by the condition, we further construct the interference UT set (IUS) and design a low-dimensional interference sparsity-aware detector (ISD) for each UT by minimizing mean-squared error (MSE), then the ISTR is developed by incorporating ISDs. Moreover, the optimality of the ISTR is proved under a relaxed version of the condition for sufficient observation selection.
- Using the structural properties of the beam matrix and the sparsity of the beam domain channel in BBCM, we propose an efficient implementation of ISTR by integrating approximate computation of the ISD, the signal reconstructed by ISD and the channel Gram matrix. Moreover, instead of using instantaneous channel state information (iCSI), we propose an efficient construction of IUS using statistical CSI (sCSI), effectively reducing the overhead for updating IUS.

E. Organization and Notations

The rest of the paper is organized as follows. The BBCM for HF skywave massive MIMO is introduced in Section II. In Section III, we discuss the sufficient observation selection and propose the ISTR. In Section IV, we propose an efficient implementation of the ISTR and an efficient construction of IUS, respectively. Sections V and VI provide simulation results and conclusions of the paper, respectively.

Notations: Vectors (matrices) are represented by Boldface lower (upper) case letters. $\mathbb{B}$, $\mathbb{R}$ and $\mathbb{C}$ denote the sets of all binary numbers, real numbers and complexity numbers, respectively, and $\mathcal{Z}_N^+ \triangleq \{1, \cdots, N\}$. The conjugate, transpose and conjugate-transpose are respectively denoted by $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$ and $(\cdot)^{\mathrm{H}}$. $\mathbf{O}$ and $\mathbf{I}_M$ indicate zero matrix and M-dimensional identity matrix, respectively. $\mathbf{0}$, $\mathbf{1}$ and $\mathbf{e}_m$ represent all-zero

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TABLE I
SYMBOL LIST

Notation	Definition
$\mathbf{H} \in \mathbb{C}^{M\times U}$	Space domain channel matrix
$\mathbf{D} \in \mathbb{C}^{A\times U}$	Beam domain channel matrix
$\mathbf{V} \in \mathbb{C}^{M\times A}$	Beam matrix
$\mathbf{G} \in \mathbb{C}^{U\times U}$	Channel Gram matrix $\mathbf{H}^{\mathrm{H}}\mathbf{H}$
$\mathbf{R}_u \in \mathbb{C}^{M\times M}$	Space domain channel covariance matrix for UT u
$\mathbf{\Omega}_u \in \mathbb{R}^{A\times A}$	Beam domain channel covariance matrix for UT u
$\boldsymbol{\omega}_u \in \mathbb{R}^{A}$	Beam domain channel coupling vector for UT u
$\mathbf{y} \in \mathbb{C}^{M}$	Received signal at the BS
$\boldsymbol{\theta} \in \mathbb{C}^{U}$	Received signal after MF
$\boldsymbol{\theta}_u \in \mathbb{C}^{N_u}$	Observation vector for the detection of x_u
$\mathbf{q}_u \in \mathbb{B}^{N}$	Bit vector mapped to symbol x_u
$\mathbf{x} \in \mathbb{C}^{U}$	Transmitted signal vector for all UTs
$\boldsymbol{\mu} \in \mathbb{C}^{U}$	A priori mean of $\mathbf{x}$
$\mathbf{\Sigma} \in \mathbb{R}^{U\times U}$	A priori variance of $\mathbf{x}$
$\mathbf{r}_u \in \mathbb{C}^{M}$	MMSE detector for UT u
$\mathbf{w}_u \in \mathbb{C}^{N_u}$	Interference sparsity-aware detector (ISD) for UT u
$L_{\mathrm{e}}(q_{u,i}\|\boldsymbol{\theta}_u)$	Extrinsic LLR of $q_{u,i}$ output by SISO detector
$L_{\mathrm{a}}(c_{u,i})$	A priori LLR of $c_{u,i}$ for SISO decoder
$L'_{\mathrm{e}}(c_{u,i})$	Extrinsic LLR of $c_{u,i}$ output by SISO decoder
$L'_{\mathrm{a}}(q_{u,i})$	A priori LLR of $q_{u,i}$ for SISO detector
$\eta_{u,u'}$	Channel correlation between UT u and u'
$\overline{\eta}_{u,u'}$	Statistical channel correlation between UT u and u'
$\mathcal{A}_u$	Non-zero beam set for UT u
$\mathcal{N}_u$	Interference UT set (IUS) for UT u

vector, all-one vector and the m-th column of identity matrix, respectively. $[\mathbf{a}]_i$ and $[\mathbf{A}]_{i,j}$ denote the i-th element of $\mathbf{a}$ and the (i,j)-th element of $\mathbf{A}$, respectively. $\bar{\jmath} = \sqrt{-1}$, $\odot$ and $|\cdot|$ denote the imaginary unit, the Hadamard product operator and the cardinality for a set or the absolute value for a scalar, respectively. $\mathbb{E}\{\cdot\}$ and $\mathrm{Cov}\{\cdot\}$ denote the ensemble expectation and covariance, respectively. $\langle\cdot\rangle_N$, $\mathrm{tr}\{\cdot\}$, and $\|\cdot\|$ represent the modulo-N operation, trace operation, and the Frobenius norm, respectively. We employ $\lfloor\cdot\rfloor$ and $\lceil\cdot\rceil$ as the floor and ceiling operations, respectively. $\mathrm{diag}\{\mathbf{a}\}$ and $\mathcal{CN}(\mathbf{a},\mathbf{A})$ respectively indicate the diagonal matrix with $\mathbf{a}$ along its main diagonal and the complex Gaussian distribution with mean $\mathbf{a}$ and covariance $\mathbf{A}$. The cyclic shift matrix and its variant are denoted by $\mathbf{\Pi}_N \triangleq [\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_N, \mathbf{e}_1] \in \mathbb{R}^{N\times N}$ and $\overline{\mathbf{\Pi}}_N \triangleq [\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_N, -\mathbf{e}_1] \in \mathbb{R}^{N\times N}$, respectively. And $\mathbf{S}(N,\mathcal{T}) \triangleq [\mathbf{e}_{n_1}, \mathbf{e}_{n_2}, \cdots, \mathbf{e}_{n_T}] \in \mathbb{R}^{N\times T}$ is a selection matrix with $\mathcal{T} = \{n_1, \cdots, n_T\} \subseteq \mathcal{Z}_N^+$, and $n_1 < \cdots < n_T$, $T = |\mathcal{T}|$. The major symbols adopted in the paper is listed in Table I for the convenience of reference.

II. SYSTEM MODEL

In this section, we present the system configuration for HF skywave massive MIMO and introduce the BBCM, which is crucial for the subsequent discussion on receiver design.

A. System Configuration

Consider a long-range HF skywave massive MIMO orthogonal frequency division multiplexing (OFDM) system with carrier frequency f_{c} and subcarrier spacing Δ_{f}. And U single-antenna UTs are served by the BS with M-antenna uniform linear array (ULA).

In HF skywave communication systems, the carrier frequency, f_{c}, needs to be adjusted when the ionospheric environment changes [44] while the inter-antenna spacing cannot be configured accordingly. To this end, different from the half wavelength of carrier in traditional terrestrial cellular massive MIMO communications, the inter-antenna spacing of HF skywave massive MIMO is typically set as $d = \lambda_{\mathrm{o}}/2$, where $\lambda_{\mathrm{o}} = c/f_{\mathrm{o}}$ with the speed of light c, and $f_{\mathrm{o}} > f_{\mathrm{c}}$ is denoted as the highest system operating frequency.[1]

B. Beam Based Channel Model

We assume the channel states stay constant in each OFDM symbol and focus on the uplink transmission over one OFDM symbol. The data transmitted by the u-th UT at the s-th subcarrier is denoted as $x_{u,s}$. Consequently, after OFDM demodulation, the signal received by the m-th antenna at the s-th subcarrier is given as

$$y_{m,s} = \sum_{u=1}^{U} h_{u,m,s} x_{u,s} + z_{m,s}, \tag{1}$$

where $z_{m,s} \sim \mathcal{CN}(0, \sigma_{\mathrm{z}})$ is the complex additive white Gaussian noise, $h_{u,m,s}$ represents the channel frequency response at the s-th subcarrier between UT u and the m-th antenna at the BS. Stacking the channel frequency responses of all antennas yields the space domain channel vector $\mathbf{h}_{u,s} \triangleq [h_{u,1,s}, \cdots, h_{u,M,s}]^{\mathrm{T}} \in \mathbb{C}^{M}$, which can be given as [11]

$$\mathbf{h}_{u,s} = \sum_{p=1}^{P_u} \alpha_{u,p} \exp\{-\bar{\jmath}2\pi s \Delta_{\mathrm{f}} \tau_{u,p}\} \mathbf{v}(\phi_{u,p}), \tag{2}$$

with the steering vector [45]

$$\mathbf{v}(\phi) \triangleq \frac{\omega(\phi)}{\sqrt{M}} [\exp\{-\bar{\jmath}2\pi\phi\}, \cdots, \exp\{-\bar{\jmath}2\pi M\phi\}]^{\mathrm{T}} \in \mathbb{C}^{M}. \tag{3}$$

In (3), $\omega(\phi) = \exp\{\bar{\jmath}\pi(M+1)\phi\}$, the scaled directional cosine $\phi = f_{\mathrm{c}}\Delta_\tau\Omega \in [-f_{\mathrm{c}}\Delta_\tau, f_{\mathrm{c}}\Delta_\tau)$, the directional cosine $\Omega \in [-1, 1)$, and $\Delta_\tau \triangleq d/c = 1/(2f_{\mathrm{o}})$. In (2), P_u is the number of paths from UT u to the BS, $\alpha_{u,p}$, $\tau_{u,p}$ and $\phi_{u,p}$ represent the complex gain, delay and scaled directional cosine of path p from UT u, respectively.

To achieve a sparse representation of HF skywave massive MIMO channels, we adopt the beam based channel model (BBCM) which has been utilized in previous works for HF skywave massive MIMO communications [6], [7], [8], [9], [10], [11], [12], [43]. Subsequently, we briefly introduce the concept of BBCM by performing uniform sampling for scaled directional cosine with deterministic sampling interval length, which is equivalent to the modified BBCM in [43] and superior to the BBCM in [6], [7], [8], [9], [10], [11], and [12] due to the fact that the proposed beam matrix has a straightforward structure related to discrete Fourier transform (DFT). Specifically, we set the interval length for uniform sampling as $\Delta_\phi = 1/S$ with $S = FM$, where F is the fine factor, and symmetric sampling is considered. Consequently, the a-th sampled scaled directional cosine is obtained as

[1]The effect of mutual coupling caused by dense inter-antenna spacing is not considered.

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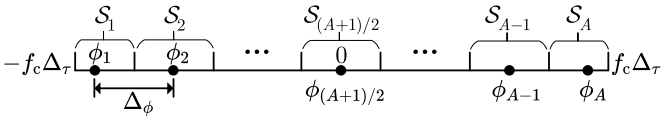


Fig. 1. Uniform sampling for scaled directional cosine.

$\phi_a = (a-1-\lfloor f_c\Delta_\tau/\Delta_\phi\rfloor)\Delta_\phi$, and the range of scaled directional cosine $[-f_c\Delta_\tau, f_c\Delta_\tau)$ can be partitioned into $A = 2\lfloor f_c\Delta_\tau/\Delta_\phi\rfloor + 1$ mutually exclusive subsets $\mathcal{S}_1, \cdots, \mathcal{S}_A$ with the a-th subset being the interval being

$$\mathcal{S}_a \triangleq \big[\max\{-f_c\Delta_\tau, \phi_a - \Delta_\phi/2\}, \min\{f_c\Delta_\tau, \phi_a + \Delta_\phi/2\}\big). \tag{4}$$

Fig. 1 visualizes the uniform sampling for scaled directional cosine.

After performing uniform sampling for scaled directional cosine, we further approximate steering vectors in (2) as sampled steering vectors and ultimately derive the BBCM. By approximating $\mathbf{v}(\phi_{u,p})$ as $\mathbf{v}(\phi_a)$, if $\phi_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_a$, with $\mathcal{P}_u \triangleq \{\phi_{u,1}, \cdots, \phi_{u,P_u}\}$, the space domain channel in (2) can be modelled as

$$\mathbf{h}_{u,s} = \sum_{a=1}^{A} \underbrace{\sum_{\phi_{u,p}\in\mathcal{P}_u\cap\mathcal{S}_a} \alpha_{u,p} \exp\{-\bar{\jmath}2\pi s\Delta_f \tau_{u,p}\}}_{=d_{u,a,s}} \mathbf{v}(\phi_a) = \mathbf{V}\mathbf{d}_{u,s}, \tag{5}$$

where $\mathbf{d}_{u,s} \triangleq [d_{u,1,s}, \cdots, d_{u,A,s}]^{\mathrm{T}} \in \mathbb{C}^A$ is the beam domain channel vector, and $\mathbf{V} \triangleq [\mathbf{v}(\phi_1), \cdots, \mathbf{v}(\phi_A)] \in \mathbb{C}^{M\times A}$ is termed as the beam matrix. Combining (3), the beam matrix can be expressed as

$$\mathbf{V} = \frac{1}{\sqrt{M}} \mathrm{diag}\{\boldsymbol{\lambda}_1\} \mathbf{I}_{M,S} \mathbf{F}_S \mathbf{I}_{S,A} \mathrm{diag}\{\boldsymbol{\lambda}_2\}, \tag{6}$$

where $\mathbf{F}_S = [\mathbf{f}_{S,1}, \cdots, \mathbf{f}_{S,S}] \in \mathbb{C}^{S\times S}$ denotes S-dimensional DFT matrix with $[\mathbf{F}_S]_{i,j} = \exp\{-\bar{\jmath}2\pi(i-1)(j-1)/S\}$, $\mathbf{I}_{N_1,N_2}$ is constructed by the first N_1 columns ($N_1 \le N_2$) of $\mathbf{I}_{N_2}$ or the first N_2 rows ($N_1 > N_2$) of $\mathbf{I}_{N_1}$, and the phase-shift vectors

$$\boldsymbol{\lambda}_1 = \mathbf{I}_{M,S} \mathbf{f}_{S,S-(A-1)/2+1}, \tag{7a}$$

$$\boldsymbol{\lambda}_2 = \mathbf{I}_{A,2S} \boldsymbol{\Pi}_{2S}^{(A-1)/2} \mathbf{f}_{2S,M}^{*}. \tag{7b}$$

According to (6), beam matrix $\mathbf{V}$ exhibits the structure of a phase-shifted partial DFT matrix. We call (5) as the BBCM. It can be written in a more compact form

$$\mathbf{H}_s = \mathbf{V}\mathbf{D}_s, \tag{8}$$

where at the s-th subcarrier, the space domain channel matrix and the beam domain channel matrix are defined as $\mathbf{H}_s \triangleq [\mathbf{h}_{1,s}, \cdots, \mathbf{h}_{U,s}] \in \mathbb{C}^{M\times U}$ and $\mathbf{D}_s \triangleq [\mathbf{d}_{1,s}, \cdots, \mathbf{d}_{U,s}] \in \mathbb{C}^{A\times U}$, respectively.

To characterize the channel statistical properties under the BBCM, the space domain and beam domain channel covariance matrices at the s-th subcarrier for UT u can be respectively defined as [9]

$$\mathbf{R}_{u,s} \triangleq \mathbb{E}\{\mathbf{h}_{u,s}\mathbf{h}_{u,s}^{\mathrm{H}}\} = \mathbf{V}\boldsymbol{\Omega}_{u,s}\mathbf{V}^{\mathrm{H}}, \tag{9a}$$

$$\boldsymbol{\Omega}_{u,s} \triangleq \mathbb{E}\{\mathbf{d}_{u,s}\mathbf{d}_{u,s}^{\mathrm{H}}\} = \mathrm{diag}\{\boldsymbol{\omega}_{u,s}\}, \tag{9b}$$

where $\boldsymbol{\omega}_{u,s} \triangleq \mathbb{E}\{\mathbf{d}_{u,s} \odot \mathbf{d}_{u,s}^{*}\} \in \mathbb{R}^A$ is the beam domain channel coupling vector with $[\boldsymbol{\omega}_{u,s}]_a = \sum_{\phi_{u,p}\in\mathcal{P}_u\cap\mathcal{S}_a} |\alpha_{u,p}|^2$, which is independent of subcarriers [6]. For brevity, we denote $\mathbf{R}_u = \mathbf{R}_{u,s}$, $\boldsymbol{\Omega}_u = \boldsymbol{\Omega}_{u,s}$ and $\boldsymbol{\omega}_u = \boldsymbol{\omega}_{u,s}$. Due to the narrow angle spread of HF skywave channels [46], the cardinality of $\mathcal{P}_u$ for each UT is limited. This fact leads to remarkable sparsity of the beam domain channel, that is, the number of non-zero elements of $\mathbf{d}_{u,s}$ and $\boldsymbol{\omega}_u$ is small. Denote the non-zero beam set as

$$\mathcal{A}_u \triangleq \{a \in \mathcal{Z}_A^+ | [\boldsymbol{\omega}_u]_a \neq 0\} \text{ with } A_u = |\mathcal{A}_u|. \tag{10}$$

Additionally, channel coupling vector $\boldsymbol{\omega}_u$ is referred to as the beam domain statistical channel state information (sCSI), whose stable period is typically longer than the channel coherent time [47]. In this paper, we assume that beam domain sCSI $\boldsymbol{\omega}_u$ and instantaneous CSI (iCSI) $\mathbf{d}_{u,s}$ are both obtained at the BS, which can be achieved by channel estimation and channel prediction methods with reliable accuracy and relatively low computational complexity and low pilot overhead [9], [10].

III. Interference Sparsity-Aware Turbo Receiver

In this section, we first introduce the principal of turbo receiver and reveal the interference sparsity in HF skywave massive MIMO systems. Then, employing the interference sparsity, we establish the selection criterion for sufficient observations without mutual information loss in SISO signal detection. Ultimately, we develop a turbo receiver design with interference sparsity.

A. Principal of Turbo Receiver

In the framework of turbo receiver, a bit-interleaved coded modulation (BICM) system is usually considered [37]. At the transmitter, the original information bitstream is sequentially encoded by an encoder and interleaved by an interleaver. After interleaving, the bitstream is partitioned into vectors of length N and mapped to symbols in constellation $\mathbb{S} = \{s_1, \cdots, s_K\}$ with modulation order $K = 2^N$, where $s_k \in \mathbb{C}$ is the k-th constellation point mapped by the bit vector $\mathbf{b}_k = [b_{k,1}, \cdots, b_{k,N}]^{\mathrm{T}} \in \mathbb{B}^N$. For UT u at the subcarrier s, a bit vector after interleaving is denoted as $\mathbf{q}_u = [q_{u,1}, \cdots, q_{u,N}]^{\mathrm{T}} \in \mathbb{B}^N$ and mapped to the transmitted symbol $x_u \in \mathbb{S}$, where index s of subcarrier is omitted for brevity. Consequently, transmitted signal vector $\mathbf{x}$ for all UTs is given by $\mathbf{x} = [x_1, \cdots, x_U]^{\mathrm{T}} \in \mathbb{S}^U$, and the received signal in (1) can be written in a compact form

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{z}, \tag{11}$$

where received signal $\mathbf{y} \triangleq [y_1, \cdots, y_M]^{\mathrm{T}} \in \mathbb{C}^M$ and noise $\mathbf{z} \triangleq [z_1, \cdots, z_M]^{\mathrm{T}} \in \mathbb{C}^M$. It is assumed that $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \sigma_z\mathbf{I})$. The a priori mean and variance of transmitted signal vector $\mathbf{x}$ are respectively denoted as $\boldsymbol{\mu} \triangleq \mathbb{E}\{\mathbf{x}\} \in \mathbb{C}^U$ and $\boldsymbol{\Sigma} \triangleq \mathrm{Cov}\{\mathbf{x}, \mathbf{x}\} \in \mathbb{R}^{U\times U}$, where $\boldsymbol{\Sigma}$ is a diagonal matrix, indicating the uncorrelatedness among the symbols transmitted by different UTs. At the BS, a turbo receiver is implemented to iteratively decode the information bits from all UTs. During the iteration, soft information of each bit is exchanged between a SISO detector and a SISO decoder. The soft information

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output by SISO detector is the extrinsic LLR of each bit, which is computed by the a priori LLR of each bit and the detection of each transmitted symbol. In what follows, we briefly introduce the procedures of SISO signal detection, mainly including the extrinsic LLR calculation and the a priori information update.

For a SISO detector, conditioned on observation vector $\mathbf{y}$, the a posteriori LLR of bit $q_{u,i}$ is given by

$$\begin{aligned} L(q_{u,i}|\mathbf{y}) &= \ln \frac{P(q_{u,i}=1|\mathbf{y})}{P(q_{u,i}=0|\mathbf{y})} = \ln \frac{\sum_{s_k\in\mathbb{S}_i^1} P(x_u=s_k|\mathbf{y})}{\sum_{s_k\in\mathbb{S}_i^0} P(x_u=s_k|\mathbf{y})} \\ &= \underbrace{\ln \frac{\sum_{s_k\in\mathbb{S}_i^1} p(\mathbf{y}|x_u=s_k)\prod_{j\neq i}^{N} P(q_{u,j}=b_{k,j})}{\sum_{s_k\in\mathbb{S}_i^0} p(\mathbf{y}|x_u=s_k)\prod_{j\neq i}^{N} P(q_{u,j}=b_{k,j})}}_{\triangleq L_{\mathrm{e}}(q_{u,i}|\mathbf{y})} \\ &\quad + \ln \frac{P(q_{u,i}=1)}{P(q_{u,i}=0)}, \end{aligned} \tag{12}$$

where $P(\cdot)$ and $p(\cdot)$ represent the probability distribution of discrete random variables and the probability density function of continuous random variables, respectively, $\mathbb{S}_i^b \triangleq \{s_k \in \mathbb{S}|b_{k,i}=b\},\ b\in\mathbb{B}, i\in\mathcal{Z}_N^+$, is the symbol set with the i-th mapping bit $b_{k,i}$ of s_k being b. The extrinsic LLR $L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$ in (12) is the output of SISO detection and used for SISO decoding.

Typically, for SISO MMSE detection, $L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$ can be obtained as [48]

$$L_{\mathrm{e}}(q_{u,i}|\mathbf{y}) = \ln \frac{\sum_{s_k\in\mathbb{S}_i^1} \exp\left\{-|\mu_u^{\mathrm{e}}-s_k|^2/\sigma_u^{\mathrm{e}} + \sum_{j\neq i}^{N}(b_{k,j}-1/2)L_{\mathrm{a}}'(q_{u,j})\right\}}{\sum_{s_k\in\mathbb{S}_i^0} \exp\left\{-|\mu_u^{\mathrm{e}}-s_k|^2/\sigma_u^{\mathrm{e}} + \sum_{j\neq i}^{N}(b_{k,j}-1/2)L_{\mathrm{a}}'(q_{u,j})\right\}}, \tag{13}$$

in which the a priori LLR $L_{\mathrm{a}}'(q_{u,i})$ is initialized by 0 for the first iteration, and updated via the soft information output by the SISO decoder for the subsequent iterations, and the extrinsic mean and the extrinsic variance of x_u are respectively given by

$$\mu_u^{\mathrm{e}} = (\widetilde{x}_u/\widetilde{\epsilon}_u - \mu_u/\sigma_u)\sigma_u^{\mathrm{e}}, \tag{14a}$$

$$\sigma_u^{\mathrm{e}} = (1/\widetilde{\epsilon}_u - 1/\sigma_u)^{-1}, \tag{14b}$$

where $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ are the signal reconstructed by the detector and the detection MSE, respectively, which are given by [48]

$$\widetilde{x}_u = \mathbf{r}_u^{\mathrm{H}}(\mathbf{y}-\mathbf{H}\boldsymbol{\mu}) + \mu_u, \tag{15a}$$

$$\widetilde{\epsilon}_u = \left(1-\mathbf{r}_u^{\mathrm{H}}\mathbf{h}_u\right)\sigma_u. \tag{15b}$$

In (15), MMSE detector $\mathbf{r}_u = (\mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^{\mathrm{H}}+\sigma_{\mathrm{z}}\mathbf{I})^{-1}\mathbf{h}_u\sigma_u$, $\mu_u = [\boldsymbol{\mu}]_u$, and $\sigma_u = [\boldsymbol{\Sigma}]_{u,u}$.

Subsequently, the a priori soft information $L_{\mathrm{a}}(c_{u,i})$ for SISO decoder is obtained by mapping $L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$ to $L_{\mathrm{a}}(c_{u,i})$ via deinterleaving $c_{u,i} = \Pi^{-1}(q_{u,i})$ with interleaver $\Pi(\cdot)$. Then, $L_{\mathrm{e}}'(c_{u,i})$ is obtained after SISO decoding, and the a priori soft information $L_{\mathrm{a}}'(q_{u,i})$ for the input of SISO detector is acquired by mapping $L_{\mathrm{e}}'(c_{u,i})$ to $L_{\mathrm{a}}'(q_{u,i})$ via interleaving $q_{u,i} = \Pi(c_{u,i})$. As a result, the probability of each information bit is calculated by

$$P(q_{u,i}=b) = \frac{1}{2}\left(1+(2b-1)\tanh\left(L_{\mathrm{a}}'(q_{u,i})/2\right)\right). \tag{16}$$

Thus the a priori mean and variance of the transmitted symbol x_u are respectively updated as

$$\mu_u = \sum_{k=1}^{K} s_k P(x_u=s_k), \tag{17a}$$

$$\sigma_u = \sum_{k=1}^{K} |s_k|^2 P(x_u=s_k) - |\mu_u|^2, \tag{17b}$$

where $P(x_u=s_k) = P(\mathbf{q}_u=\mathbf{b}_k) = \prod_{i=1}^{N} P(q_{u,i}=b_{k,i})$. To improve the BER performance, the updated a priori information is used for the next iteration of signal detection, i.e., the calculations of $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ for each UT u.

B. Sufficient Observation for SISO Signal Detection With Interference Sparsity

In the subsequent discussion, we focus primarily on SISO signal detection in the turbo receiver. The goal of signal detection is to recover $\mathbf{x}$ based on observation $\mathbf{y}$, channel $\mathbf{H}$ and a priori statistics $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. When the average power of each transmitted symbol is normalized to unit, the values of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are initialized as $\mathbf{0}$ and $\mathbf{I}_U$, respectively, and are updated by (17) for SISO signal detection.

However, for massive MIMO communications, when the number of antennas at the BS is large, the dimensionality of $\mathbf{y}$ becomes high for directly performing signal detection, which brings high-dimensional matrix operations and thus leads to unbearable complexity. Therefore, we try to find a low-dimensional observation vector for signal detection of each UT. Before proceeding, we adopt the following definition to measure the sufficiency of observation vectors.

Definition 1: Vector $\boldsymbol{\theta}_u$ is defined as a sufficient observation for the detection of x_u if

$$I(x_u, \boldsymbol{\theta}_u) = I(x_u, \mathbf{y}), \tag{18}$$

where $I(\cdot,\cdot)$ is mutual information between random variables.

From Definition 1, for the detection of x_u, observation $\boldsymbol{\theta}_u$ is sufficient if it retains the mutual information with x_u compared to the original observation $\mathbf{y}$.

A well-known sufficient observation, also called sufficient statistic, for the detection of $\mathbf{x}$ is the received signal after MF [49]

$$\boldsymbol{\theta} = \mathbf{H}^{\mathrm{H}}\mathbf{y} = \mathbf{G}\mathbf{x} + \mathbf{n}, \tag{19}$$

where $\mathbf{G} \triangleq \mathbf{H}^{\mathrm{H}}\mathbf{H} \in \mathbb{C}^{U\times U}$, and $\mathbf{n} \triangleq \mathbf{H}^{\mathrm{H}}\mathbf{z} \in \mathbb{C}^{U}$. The (i,j)-th element of channel Gram matrix $\mathbf{G}$ is the inner product $g_{i,j} = \mathbf{h}_i^{\mathrm{H}}\mathbf{h}_j$. For $i\neq j$, the value of $|g_{i,j}|$ indicates the interference level between UT i and UT j [50]. If $\mathbf{h}_i$ and $\mathbf{h}_j$ are orthogonal, namely $g_{i,j}=0$, then there is no interference between UT i and UT j. Building on BBCM as in (5) and the sparsity of beam domain channels, the space domain channel for UT u can be expressed as $\mathbf{h}_u = \sum_{a\in\mathcal{A}_u} d_{u,a}\mathbf{v}(\phi_a)$, then

$$\begin{aligned} g_{i,j} &= \sum_{a\in\mathcal{A}_i}\sum_{a'\in\mathcal{A}_j} d_{i,a}^* d_{j,a'}\mathbf{v}^{\mathrm{H}}(\phi_a)\mathbf{v}(\phi_{a'}) \\ &= \sum_{a\in\mathcal{A}_i}\sum_{a'\in\mathcal{A}_j} d_{i,a}^* d_{j,a'}\gamma_{|a-a'|}, \end{aligned} \tag{20}$$

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where $\gamma_{|a-a'|} = \mathbf{v}^{\mathrm{H}}(\phi_a)\mathbf{v}(\phi_{a'})$, and

$$\gamma_n = \begin{cases} \sin(\pi n/F)/(M\sin(\pi n/S)), & \langle n\rangle_S \neq 0 \\ 1, & \langle n\rangle_S = 0 \end{cases}, \tag{21}$$

is a sinc-like sequence. Typically, most values of γ_n are close to zero, especially when M is sufficiently large and $n \gg 1$. For HF skywave massive MIMO scenarios, due to the fact that the geographical distances between most UTs are substantially large, the non-zero beam sets for different UTs are usually well separated in practice, i.e., $|\phi_a - \phi_{a'}|$ is large for $a \in \mathcal{A}_i$ and $a' \in \mathcal{A}_j$ with most $i \neq j$. Therefore, for most pairs (i,j) of UTs, the values of $|a-a'|$ for $a \in \mathcal{A}_i$ and $a' \in \mathcal{A}_j$ are usually large, which leads to small values of $\gamma_{|a-a'|}$. Moreover, the cardinalities of $\mathcal{A}_i$ and $\mathcal{A}_j$ are small due to the limited angle spread of each UT [6]. As a result, the interference terms $g_{i,j}$ for most pairs (i,j) of UTs are close to zero as per (20), which leads to the numerical sparsity of $\mathbf{G}$. Hereafter, we refer to such sparsity as *interference sparsity*, which is used for observation selection from the output of MF.

Next, we extract elements from $\boldsymbol{\theta}$ to construct an observation vector $\boldsymbol{\theta}_u = \mathbf{N}_u^{\mathrm{T}}\boldsymbol{\theta}$ for the detection of each symbol x_u, where $\mathbf{N}_u \triangleq \mathbf{S}(U,\mathcal{N}_u) \in \mathbb{R}^{U\times N_u}$ and $\mathcal{N}_u \subseteq \mathcal{Z}_U^+$ are respectively UT selection matrix and UT set, and $N_u = |\mathcal{N}_u|$. In order to construct a sufficient observation $\boldsymbol{\theta}_u$ for the detection of each transmitted symbol x_u, by making the assumption $\mathbf{x} \sim \mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, we have the following theorem and corollary, where the proof of Theorem 1 is provided in Appendix A.

Theorem 1: For a given observation vector $\boldsymbol{\theta}_u$,

$$I(x_u, \boldsymbol{\theta}_u) \leq I(x_u, \mathbf{y}), \tag{22}$$

where the equality holds when $u \in \mathcal{N}_u$ and

$$g_{i,j} = 0, \ \forall i \in \mathcal{N}_u, j \notin \mathcal{N}_u. \tag{23}$$

Corollary 1: If $g_{i,j} = 0, \ \forall i \neq j$, then $\forall u \in \mathcal{Z}_U^+$,

$$I(x_u, \theta_u) = I(x_u, \mathbf{y}), \tag{24}$$

where $\theta_u = [\boldsymbol{\theta}]_u$.

As shown above, the orthogonality condition (23), the channels of UTs in a set containing UT u are orthogonal to those of UTs outside the set, can be asymptotically achieved when the number M of antennas is large, i.e., the interference sparsity. Thus, Theorem 1 basically says that with the interference sparsity, selecting the subvector of $\boldsymbol{\theta}$ corresponding to this UT set forms a sufficient observation for the detection of x_u. Corollary 1 can be directly obtained by setting $\mathbf{N}_u = \mathbf{e}_u$ in Theorem 1. From Corollary 1, if $\mathbf{G}$ is a diagonal matrix, for any UT u, scalar θ_u is a sufficient observation for the detection of x_u. In this special case, the channels between any pair of UTs are orthogonal, which can be interpreted as the absence of interference among UTs.

C. Interference Sparsity-Aware Turbo Receiver

According to III-B, we choose observation vector $\boldsymbol{\theta}_u$ for the detection of x_u with interference sparsity. To determine an effective metric of interference level between any pair of UTs, we denote channel correlation between UT i and j by normalizing $|g_{i,j}|$ as

$$\eta_{i,j} \triangleq \frac{\left|\mathbf{h}_i^{\mathrm{H}}\mathbf{h}_j\right|}{\|\mathbf{h}_i\|\|\mathbf{h}_j\|} = \frac{|g_{i,j}|}{\sqrt{g_{i,i}g_{j,j}}}. \tag{25}$$

Clearly, $0 \leq \eta_{i,j} \leq 1$, $\eta_{i,j} = \eta_{j,i}$, and $\eta_{i,j} = 0$ when $\mathbf{h}_i$ and $\mathbf{h}_j$ are orthogonal, $\eta_{i,j} = 1$ when $\mathbf{h}_i$ is a scaled version of $\mathbf{h}_j$. Therefore, the channel correlation $\eta_{i,j}$ can serve as a metric of the interference level among UTs. A larger value of $\eta_{i,j}$ implies greater interference among UT i and UT j, and vice versa. Theorem 1 implies that the detection of x_u can be achieved without mutual information loss by only considering the interference from $\mathcal{N}_u$ if the condition (23) is satisfied. As mentioned earlier, with the interference sparsity, (23) approximately holds and thus we can construct a UT set with strong interference to UT u, and determine the observation for signal detection by the UT set. Specifically, we define the interference UT set (IUS) for UT u as

$$\mathcal{N}_u \triangleq \left\{u' \in \mathcal{Z}_U^+ \middle| \eta_{u,u'} > \eta_{\mathrm{th}}\right\} \text{ with } N_u = |\mathcal{N}_u|, \tag{26}$$

where $\eta_{\mathrm{th}} < 1$ is a preset threshold. Admittedly, the equality condition in Theorem 1 may not be strictly satisfied with the IUS constructed by (26). However, it can be verified by simulation that the gap between $I(x_u, \boldsymbol{\theta}_u)$ and $I(x_u, \mathbf{y})$ is very small in this case. Therefore, we can still construct an observation for the detection of x_u without significant loss of sufficiency by selecting the subvector of $\boldsymbol{\theta}$ corresponding to the IUS as $\boldsymbol{\theta}_u = \mathbf{N}_u^{\mathrm{T}}\boldsymbol{\theta}$, which can be expressed as

$$\boldsymbol{\theta}_u = \mathbf{G}_u\mathbf{x} + \mathbf{n}_u, \tag{27}$$

where $\mathbf{G}_u = \mathbf{N}_u^{\mathrm{T}}\mathbf{G}$, $\mathbf{n}_u = \mathbf{N}_u^{\mathrm{T}}\mathbf{n}$. We next demonstrate the influence of enlarging the IUS. For a given IUS $\mathcal{N}_u$ of UT u, we construct an expanded IUS $\mathcal{N}_u' \supseteq \mathcal{N}_u$ by adding more UTs into $\mathcal{N}_u$. Then the observation vector based on $\mathcal{N}_u'$ is expressed as $\boldsymbol{\theta}_u' = \left(\mathbf{N}_u'\right)^{\mathrm{T}}\boldsymbol{\theta}$, where $\mathbf{N}_u' = \mathbf{S}(U,\mathcal{N}_u') \in \mathbb{R}^{U\times N_u'}$ is the UT selection matrix with $N_u' = |\mathcal{N}_u'|$. Since $\mathcal{N}_u' \supseteq \mathcal{N}_u$, then $\mathbf{N}_u$ lies on the column space of $\mathbf{N}_u'$ and $\mathbf{N}_u = \mathbf{N}_u'\left(\mathbf{N}_u'\right)^{\mathrm{T}}\mathbf{N}_u$. Therefore, we have

$$\boldsymbol{\theta}_u = \mathbf{N}_u^{\mathrm{T}}\boldsymbol{\theta} = \mathbf{N}_u^{\mathrm{T}}\mathbf{N}_u'\left(\mathbf{N}_u'\right)^{\mathrm{T}}\boldsymbol{\theta} = \mathbf{N}_u^{\mathrm{T}}\mathbf{N}_u'\boldsymbol{\theta}_u'. \tag{28}$$

According to the data processing inequality [51], we have

$$I(x_u, \boldsymbol{\theta}_u) \leq I(x_u, \boldsymbol{\theta}_u') \leq I(x_u, \boldsymbol{\theta}). \tag{29}$$

Eq. (29) indicates that expanding the IUS can increase the mutual information between the transmitted symbol and the observation vector, which is expected to improve the performance of signal detection.

After determining observation vector $\boldsymbol{\theta}_u$, we attain the detection of symbol x_u by linear operation

$$\widehat{x}_u = \mathbf{w}_u^{\mathrm{H}}\boldsymbol{\theta}_u + q_u. \tag{30}$$

Given that observation $\boldsymbol{\theta}_u$ is constructed using the interference sparsity, we refer to (30) as interference sparsity-aware signal detection for UT u. To obtain $\mathbf{w}_u$ and q_u in (30), we resort to minimizing the following MSE:

$$\widehat{\epsilon}_u \triangleq \mathbb{E}\left\{|x_u - \widehat{x}_u|^2\right\} = \mathbb{E}\left\{\left|x_u - \left(\mathbf{w}_u^{\mathrm{H}}\boldsymbol{\theta}_u + q_u\right)\right|^2\right\}. \tag{31}$$

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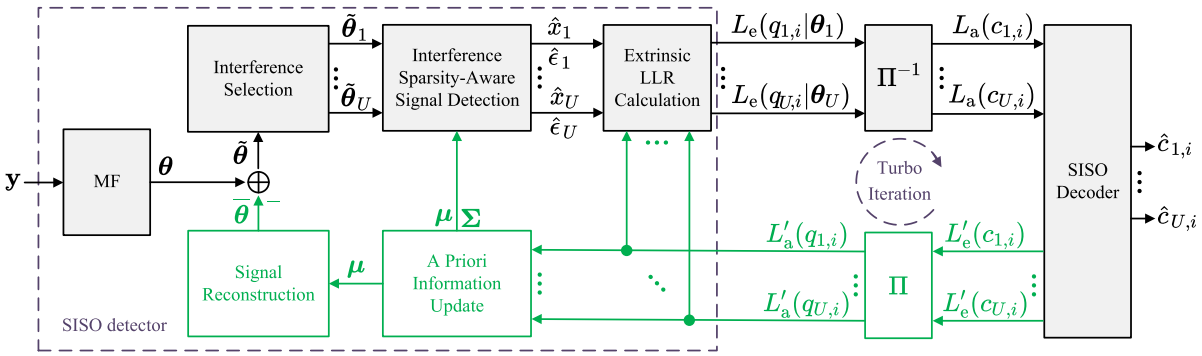


Fig. 2. Block diagram of the ISTR.

Subsequently, $\mathbf{w}_u$ and q_u can be obtained as [28]

$$\begin{aligned}\mathbf{w}_u &= \mathrm{Cov}\{\boldsymbol{\theta}_u,\boldsymbol{\theta}_u\}^{-1}\mathrm{Cov}\{\boldsymbol{\theta}_u,x_u\}\\ &= \left(\mathbf{G}_u\boldsymbol{\Sigma}\mathbf{G}_u^{\mathrm{H}}+\sigma_{\mathrm{z}}\mathbf{G}_u'\right)^{-1}\mathbf{G}_u\mathbf{e}_u\sigma_u, \qquad (32a)\\ q_u &= \mathbb{E}\{x_u\}-\mathrm{Cov}\{x_u,\boldsymbol{\theta}_u\}\mathrm{Cov}\{\boldsymbol{\theta}_u,\boldsymbol{\theta}_u\}^{-1}\mathbb{E}\{\boldsymbol{\theta}_u\}\\ &= \mu_u-\mathbf{w}_u^{\mathrm{H}}\mathbf{G}_u\boldsymbol{\mu}, \qquad (32b)\end{aligned}$$

respectively, where $\mathbf{G}_u'=\mathbf{N}_u^{\mathrm{T}}\mathbf{G}\mathbf{N}_u$ and $\mathbf{w}_u$ is termed as the interference sparsity-aware detector (ISD) for UT u. Substituting (32) into (30) and (31) yields the signal reconstructed by ISD and the detection MSE for UT u:

$$\widehat{x}_u=\mathbf{w}_u^{\mathrm{H}}\widetilde{\boldsymbol{\theta}}_u+\mu_u, \qquad (33a)$$

$$\widehat{\epsilon}_u=\left(1-\mathbf{w}_u^{\mathrm{H}}\mathbf{G}_u\mathbf{e}_u\right)\sigma_u, \qquad (33b)$$

where $\widetilde{\boldsymbol{\theta}}_u=\mathbf{N}_u^{\mathrm{T}}\widetilde{\boldsymbol{\theta}}$ is a subvector of $\widetilde{\boldsymbol{\theta}}=\boldsymbol{\theta}-\overline{\boldsymbol{\theta}}$ with $\overline{\boldsymbol{\theta}}=\mathbb{E}\{\boldsymbol{\theta}\}=\mathbf{G}\boldsymbol{\mu}$. Compared with the MMSE detection in (15), the dimensionality of ISD $\mathbf{w}_u$ is typically much lower than that of the MMSE detector $\mathbf{r}_u$, which holds the potential to lower computational complexity.

We next incorporate the ISD into the framework of turbo receiver, forming the interference sparsity-aware turbo receiver (ISTR). Using the ISD, the extrinsic information for SISO detection in turbo receiver can be obtained through (13) by replacing $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ with $\widehat{x}_u$ and $\widehat{\epsilon}_u$, respectively, yielding extrinsic LLR $L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u)$, which is conditioned on the low-dimensional observation $\boldsymbol{\theta}_u$ rather than the original observation $\mathbf{y}$. Other procedures are consistent with those discussed in III-A, then the ISTR is formulated.

In order to facilitate a clearer comprehension of the workflow of the ISTR, a block diagram of the ISTR is concluded in Fig. 2. Before the turbo iteration, the module of *MF* outputs $\boldsymbol{\theta}=\mathbf{H}^{\mathrm{H}}\mathbf{y}$. During the turbo iteration, $\widetilde{\boldsymbol{\theta}}=\boldsymbol{\theta}-\overline{\boldsymbol{\theta}}$ is first computed, then the module of *Interference Selection* extracts $\widetilde{\boldsymbol{\theta}}_u=\mathbf{N}_u^{\mathrm{T}}\widetilde{\boldsymbol{\theta}}$, $u\in\mathcal{Z}_U^+$, from $\widetilde{\boldsymbol{\theta}}$ using the IUS, and the module of *Interference Sparsity-Aware Signal Detection* designs ISDs and computes (33). Next, the extrinsic LLRs are obtained by the module of *Extrinsic LLR Calculation*. After deinterleaving, SISO decoding and interleaving, the a priori LLRs are obtained and used for SISO detection at the next iteration, where the module of *A Priori Information Update* calculates the a priori statistics through (17), and the module of *Signal Reconstruction* executes the computation of $\overline{\boldsymbol{\theta}}=\mathbf{G}\boldsymbol{\mu}$. The green part in Fig. 2 represents the relevant modules for the a priori information feedback from the SISO decoder, and $\widehat{c}_{u,i}$ is the decoded information bit output by the SISO decoder at the final iteration.

Although Theorem 1 guarantees the sufficiency of the low-dimensional observation under specific interference sparsity condition, the optimality of the derived ISTR requires further justification. The following theorem explicitly establishes the conditions under which the ISTR attains the optimality, which is proved in Appendix B.

Theorem 2: For $u\in\mathcal{N}_u$ and $\forall l\notin\mathcal{N}_u$, if one of the following conditions is satisfied:

1) $g_{k,l}=0,\ \forall k\in\mathcal{N}_u$,
2) $\sigma_l=0$,

we have

$$L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u)=L_{\mathrm{e}}(q_{u,i}|\mathbf{y}). \qquad (34)$$

From Theorem 2, under certain circumstances, the extrinsic LLR of $q_{u,i}$ conditioned on $\boldsymbol{\theta}_u$ is equal to that conditioned on $\mathbf{y}$, which reveals the optimality of the ISTR. Interestingly, the condition given by Theorem 2 is a relaxed version of the condition for the equality case of (22) in Theorem 1. Specifically, the condition stated in Theorem 2 can be interpreted as follows: for any UT beyond the IUS of UT u, either its channel is orthogonal to those of the UTs within the IUS, or the a priori variance of its transmitted symbol is zero. This indicates that, when there are UTs with the a priori variances of transmitted symbols being zero outside the IUS of a specific UT, even if the interference sparsity condition for the observation selection as described in Theorem 1 is not strictly satisfied, there will be no loss of detection performance for this UT. Fortunately, with the help of the SISO decoder in turbo receiver, the a priori information for signal detection is highly reliable after a few iterations, then most of the updated variances are close to zero [29].[2] Therefore, even if the interference sparsity condition cannot be strictly met in practice, the resulting performance degradation will be compensated for through iterations.

[2]Since the updated variance of each symbol is not exactly equal to zero during early turbo iterations, the conditions in Theorem 2 may only be partially met. Nevertheless, the updated variances are small enough to guarantee that the gap between $L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u)$ and $L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$ approximately vanishes after a few iterations. It can be concluded from the simulation results that when the updated variances are on the order of 10^{-2} or lower, the approximation is sufficiently accurate, where the required number of iterations is about 3.

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As a particular case, for linear detection, the feedback of the a priori information is unavailable, in which case the mean and variance of $\mathbf{x}$ are commonly set as $\boldsymbol{\mu} = \mathbf{0}$ and $\boldsymbol{\Sigma} = \mathbf{I}_U$, respectively. Then the detection of x_u is reduced to $\widehat{x}_u = \mathbf{w}_u^{\mathrm{H}}\boldsymbol{\theta}_u$ with $\mathbf{w}_u = \left(\mathbf{G}_u\mathbf{G}_u^{\mathrm{H}} + \sigma_{\mathrm{z}}\mathbf{G}_u'\right)^{-1}\mathbf{G}_u\mathbf{e}_u$. Specifically, some special cases are listed as below:

- If $\mathcal{N}_u = \mathcal{Z}_U^+,\ \forall u \in \mathcal{Z}_U^+$, then (33) is equivalent to MMSE detection with $\mathbf{w}_u = (\mathbf{G} + \sigma_{\mathrm{z}}\mathbf{I})^{-1}\mathbf{e}_u$ and $\boldsymbol{\theta}_u = \boldsymbol{\theta}$, in which case all other UTs are considered as interference for the detection of each UT. This case corresponds to multi-user detection.
- If $\mathcal{N}_u = \{u\},\ \forall u \in \mathcal{Z}_U^+$, then (33) is equivalent to scaled MF with a scalar detector $w_u = g_{u,u}/\left(\sum_{u'=1}^{U}|g_{u,u'}|^2 + \sigma_{\mathrm{z}}g_{u,u}\right)$ and a scalar observation $\theta_u = [\boldsymbol{\theta}]_u$, in which case no other UTs are considered as interference for the detection of each UT. This case corresponds to single-user detection.

Remark 1: We present a comparison between the proposed ISTR and BSTR [43]. While the BBCM and the framework of turbo receiver are both incorporated into the ISTR and the BSTR, the ISTR provides more profound insights and delivers notable advancements. Unlike the BSTR, which justifies the beam structured nature of space domain detectors using the beam domain channel sparsity, the ISTR employs the property of interference sparsity and conducts its reasoning from the perspectives of extracting observations after MF and mutual information loss, making it more intuitive and fundamental. Besides, the ISTR shows excellent generality since it will be reduced to certain traditional receivers in some special cases. Furthermore, since the IUS for each UT can be flexibly constructed with appropriate channel correlation threshold in the ISTR and the size of the IUS is typically much smaller than the size of the beam set for each UT group in the BSTR, the ISTR generally possesses higer design flexibility and lower computational complexity.

IV. Efficient Implementation

In this section, we develop an efficient implementation of ISTR using the specific structure of the beam matrix and various approximate computations. Further, an efficient construction of IUS using sCSI is proposed.

A. Efficient Implementation of ISTR

In this subsection, we discuss the efficient implementation of ISTR with the aid of BBCM. To begin with, we present the inherent structure of the beam matrix in BBCM and consider the case of even M, while the odd case follows a similar form. In detail, the beam matrix in (6) can be rewritten as

$$\mathbf{V} = -\frac{1}{\sqrt{M}}\mathbf{I}_{M,S}\mathbf{F}_S\boldsymbol{\Lambda}\overline{\boldsymbol{\Pi}}_S^{S-(A-1)/2}\mathbf{I}_{S,A}, \tag{35}$$

where $\boldsymbol{\Lambda} = \mathrm{diag}\left\{\mathbf{I}_{S,2S}\mathbf{f}_{2S,M}^*\right\}$. The structure of $\mathbf{V}$ in (35) is employed for efficient implementation of the ISTR. We adopt complex multiplications (CMs) as the indicator of computational complexity [52], where the multiplication of a real-valued scalar and a complex-valued scalar is regarded as half CM, and the multiplication of real-valued scalars is viewed as one-quarter of a CM. For the convenience of complexity analysis, we denote average values $\widetilde{A} = \frac{1}{U}\sum_{u=1}^{U}A_u$ and $\widetilde{N} = \frac{1}{U}\sum_{u=1}^{U}N_u$ as the average number of non-zero beams and the average number of interfering UTs, respectively. In the following, we elaborate on the efficient implementation of ISTR from three aspects: low-complexity computation of the ISD, low-complexity computation of the interference sparsity-aware signal detection, and low-complexity computation of the channel Gram matrix, along with their complexity analysis.

Firstly, we discuss the low-complexity computation of ISD $\mathbf{w}_u$. According to the construction principle of IUS, the interference from UTs outside $\mathcal{N}_u$ to those within $\mathcal{N}_u$ is small. Consequently, we can approximate that $g_{i,j} \approx 0,\ i \in \mathcal{N}_u, j \notin \mathcal{N}_u$. Thus observation $\boldsymbol{\theta}_u$ in (27) can be approximated by

$$\boldsymbol{\theta}_u = \mathbf{G}_u'\mathbf{x}_u + \sum_{u' \notin \mathcal{N}_u}\mathbf{G}_u\mathbf{e}_{u'}x_{u'} + \mathbf{n}_u \approx \mathbf{G}_u'\mathbf{x}_u + \mathbf{n}_u, \tag{36}$$

where $\mathbf{x}_u = \mathbf{N}_u^{\mathrm{T}}\mathbf{x}$ is the signal from IUS $\mathcal{N}_u$. In line with (32), $\mathbf{w}_u$ can be approximated as

$$\begin{aligned}\mathbf{w}_u &\approx \left(\mathbf{G}_u'\boldsymbol{\Sigma}_u'\mathbf{G}_u' + \sigma_{\mathrm{z}}\mathbf{G}_u'\right)^{-1}\mathbf{G}_u\mathbf{e}_u\sigma_u \\ &= \left(\boldsymbol{\Sigma}_u'\mathbf{G}_u' + \sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u\sigma_u,\end{aligned} \tag{37}$$

where $\boldsymbol{\Sigma}_u' = \mathbf{N}_u^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{N}_u$. Moreover, we approximate the a priori variance $\boldsymbol{\Sigma}$ as $\sigma\mathbf{I}$ [28], [48], where $\sigma = \mathrm{tr}\{\boldsymbol{\Sigma}\}/U$ is the average variance. Then $\mathbf{w}_u$ can be further approximated as

$$\mathbf{w}_u \approx \left(\mathbf{G}_u' + q\mathbf{I}\right)^{-1}\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u, \tag{38}$$

where $q = \sigma_{\mathrm{z}}/\sigma$. Eq. (38) avoids a series of matrix multiplications, effectively reducing the computational complexity. Notably, with well designed user schedulers, channel Gram matrix $\mathbf{G}$ can be a well-conditioned positive definite matrix, then matrix $\mathbf{G}_u' + q\mathbf{I}$ can be well-conditioned as well, even in high SNR scenarios. This implies that the numerical stability of matrix inversion in (38) can be guaranteed. Once $\mathbf{G}$ is obtained, the number of CMs of calculating $\mathbf{w}_u,\ u \in \mathcal{Z}_U^+$, is $N_{\mathbf{w}} = U\left(\widetilde{N}^3/2 + 3\widetilde{N}^2/2\right)$.

Next, we turn to the low-complexity computation of $\widehat{x}_u = \mathbf{w}_u^{\mathrm{H}}\mathbf{N}_u^{\mathrm{T}}\widetilde{\boldsymbol{\theta}} + \mu_u$ and detection MSE $\widehat{\epsilon}_u$, where the computation of $\widetilde{\boldsymbol{\theta}} = \boldsymbol{\theta} - \overline{\boldsymbol{\theta}}$ can be divided into the computation of MF $\boldsymbol{\theta} = \mathbf{H}^{\mathrm{H}}\mathbf{y}$ and signal reconstruction $\overline{\boldsymbol{\theta}} = \mathbf{G}\boldsymbol{\mu}$, respectively. Using BBCM (8) and the structure (35) of the beam matrix, the process of MF can be expressed as

$$\boldsymbol{\theta} = \mathbf{D}^{\mathrm{H}}\mathbf{V}^{\mathrm{H}}\mathbf{y} = \frac{1}{\sqrt{M}}\mathbf{D}^{\mathrm{H}}\mathbf{I}_{A,S}\overline{\boldsymbol{\Pi}}_S^{(A-1)/2}\boldsymbol{\Lambda}^*\mathbf{F}_S^{\mathrm{H}}\mathbf{I}_{S,M}\mathbf{y}. \tag{39}$$

Observing (39), the computational complexity of MF mainly comes from matrix-vector multiplications by $\mathbf{F}_S$ and $\mathbf{D}$, where the computation of $\mathbf{F}_S^{\mathrm{H}}\mathbf{a}$ can be fast implemented by inverse fast Fourier transform (IFFT), and the computation of $\mathbf{D}^{\mathrm{H}}\mathbf{b}$ only involves a small number of scalar multiplications since beam domain channel matrix $\mathbf{D}$ exhibits significant sparsity, then the complexity of MF is $N_{\boldsymbol{\theta}} = (S\log S)/2 + A + \widetilde{A}U$. The efficient implementation of MF is illustrated in Fig. 3. Since the a priori mean $\boldsymbol{\mu}$ is updated at each turbo iteration, then $\overline{\boldsymbol{\theta}}$ is updated accordingly, which can be approximately

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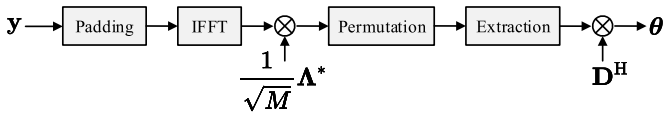


Fig. 3. Efficient implementation of MF.

computed using the asymptotic sparsity of $\mathbf{G}$. Hence, signal reconstruction requires $N_{\overline{\boldsymbol{\theta}}} = G_{\rm nz}$ CMs, where $G_{\rm nz}$ is the number of non-zero elements of approximately computed $\mathbf{G}$, which will be discussed later. After obtaining $\mathbf{w}_u$ and $\widetilde{\boldsymbol{\theta}}$, the calculation of $\widehat{x}_u, \widehat{\epsilon}_u,\ u \in \mathcal{Z}_U^+$, involves $N_{\rm det} = 2\widetilde{N}U$ CMs.

According to the above discussion, both the computation of the ISD and the computation of the signal reconstructed by ISD require channel Gram matrix $\mathbf{G} = \mathbf{H}^{\rm H}\mathbf{H}$. Then, we focus on the low-complexity computation of $\mathbf{G}$. It is assumed that accurate beam domain channel $\mathbf{D}$ is estimated for signal detection, then $\mathbf{G}$ can be computed by $\mathbf{G} = \mathbf{D}^{\rm H}\mathbf{Q}\mathbf{D}$ with $\mathbf{Q} \triangleq \mathbf{V}^{\rm H}\mathbf{V} \in \mathbb{C}^{A\times A}$. The following lemma, proved in Appendix C, reveals the inherent structure of $\mathbf{G}$, which is useful for the low-complexity computation of $\mathbf{G}$.

Lemma 1: $\mathbf{Q}$ is a real-valued symmetric Toeplitz matrix, which can be expressed as

$$\mathbf{Q} = \mathbf{I}_A + \mathbf{I}_{A,S} \sum_{n=1}^{\lceil S/2\rceil-1} \left(\overline{\mathbf{\Pi}}_S^n - \overline{\mathbf{\Pi}}_S^{S-n}\right) \mathbf{I}_{S,A}\gamma_n, \tag{40}$$

where γ_n is defined by (21).

In accordance with Lemma 1, the computation of $\mathbf{Q}\mathbf{D}$ can be effectively implemented by using the structural property of $\mathbf{Q}$ and the inherent sparsity of $\mathbf{D}$, which can be expressed as

$$\mathbf{Q}\mathbf{D} = \mathbf{D} + \mathbf{I}_{A,S} \sum_{n=1}^{\lceil S/2\rceil-1} \left(\overline{\mathbf{\Pi}}_S^n - \overline{\mathbf{\Pi}}_S^{S-n}\right) \mathbf{I}_{S,A}\mathbf{D}\gamma_n. \tag{41}$$

From (41), the computational complexity of $\mathbf{Q}\mathbf{D}$ arises from the computation of $\mathbf{D}\gamma_n$, only involving the multiplications of real-valued scalars γ_n and limited non-zero complex-valued elements in $\mathbf{D}$. As the number of antennas at the BS increases, S becomes huge, leading to a significant increase of terms of γ_n. To further reduce the computational complexity, a certain number of coefficients γ_n need to be filtered out. If fine factor F is an integer, then $\gamma_n = 0$ for $n \neq 0$, $\langle n\rangle_F = 0$. Specially, when $F = 1$, $\gamma_n = 0$ for $n \neq 0$, in which case $\mathbf{Q} = \mathbf{I}_A$ and there is no need to compute $\mathbf{Q}\mathbf{D}$. However, the sparsity of $\mathbf{D}$ deteriorates and the accurate channel becomes difficult to estimate when $F = 1$ [9]. When $F > 1$, about $1/F$ of the terms of γ_n equal 0 and $\gamma_n \ll \gamma_0 = 1$ for most n. To this end, we define a filtering set

$$\mathcal{Q} \triangleq \left\{ n \in \mathcal{Z}^+_{\lceil S/2\rceil-1} \middle| |\gamma_n| > \varepsilon \right\} \text{ with } Q = |\mathcal{Q}|, \tag{42}$$

where ε is a predefined threshold. Therefore, the computation of $\mathbf{Q}\mathbf{D}$ can be approximated by discarding the summation terms with $n \notin \mathcal{Q}$ in (41), i.e.,

$$\mathbf{Q}\mathbf{D} \approx \mathbf{P} = \mathbf{D} + \mathbf{I}_{A,S} \sum_{n\in\mathcal{Q}} \left(\overline{\mathbf{\Pi}}_S^n - \overline{\mathbf{\Pi}}_S^{S-n}\right) \mathbf{I}_{S,A}\mathbf{D}\gamma_n. \tag{43}$$

Once $\mathbf{P}$ is calculated, $\mathbf{G}$ can be approximately computed via $\mathbf{G} \approx \mathbf{D}^{\rm H}\mathbf{P}$ using its Hermitian structure and the sparsity of $\mathbf{D}$ and $\mathbf{P}$. Leveraging (43), it can be checked that $\left[\mathbf{D}^{\rm H}\mathbf{P}\right]_{i,j} = \left[\mathbf{D}^{\rm H}\mathbf{P}\right]_{j,i} = 0$ if $(i,j) \notin \mathcal{G}$, where the index set

$$\mathcal{G} = \Big\{ (i,j) \in \mathcal{Z}_U^+ \times \mathcal{Z}_U^+ \Big| (\mathcal{A}_i \cap \mathcal{A}_j) \cup (\mathcal{A}_i \cap \mathcal{A}_{j,n}) \cup (\mathcal{A}_i \cap \mathcal{A}_{j,-n}) \neq \varnothing,\ \forall n \in \mathcal{Q} \Big\}. \tag{44}$$

In (44), the non-zero beam set $\mathcal{A}_i = \{a_{i,1}, \cdots, a_{i,A_i}\}$ and its shifted version $\mathcal{A}_{i,n} = \{\langle a_{i,1}+n-1\rangle_S+1, \cdots, \langle a_{i,A_i}+n-1\rangle_S+1\}$. Thus the non-zero elements that need to be calculated in $\mathbf{G}$ can be pre-determined by $\mathcal{G}$ with number $G_{\rm nz} = |\mathcal{G}|$. Therefore, the computation of $\mathbf{D}^{\rm H}\mathbf{P}$ consumes $\widetilde{A}(G_{\rm nz}+U)/2$ CMs. Accordingly, the total CMs required for the approximate computation of $\mathbf{G}$ is $N_{\mathbf{G}} = \widetilde{A}(G_{\rm nz} + U(Q+1))/2$.

Fig. 4 illustrates the normalized magnitude of $\mathbf{G}$ in dB with $M = 256$ and $U = 72$. It can be observed that compared with the exactly computed $\mathbf{G}$ with $\varepsilon = 0$, for $\varepsilon = 0.01$ or $\varepsilon = 0.1$, the elements with relatively high amplitudes in $\mathbf{G}$ are retained while most of the elements with extremely low amplitudes are set to zero due to the approximation (43). Moreover, the larger ε leads to the smaller cardinalities of sets $\mathcal{Q}$ and $\mathcal{G}$, resulting in the more pronounced sparsity and lower computational complexity of $\mathbf{G}$. However, the approximation error of $\mathbf{G}$ also increases as ε gets larger, which may affects the performance of ISTR. To evaluate the approximation error of $\mathbf{G}$, Fig. 5 presents the normalized MSE in dB of computing $\mathbf{G}$ against the cardinality of filtering set $\mathcal{Q}$, namely, the value of Q. From Fig. 5, when $\varepsilon \leq 0.01$, or equivalently, $Q \geq 33$, the NMSE of computing $\mathbf{G}$ is lower than -30 dB, demonstrating the accuracy of the approximate computation of $\mathbf{G}$. This is because the filtered elements of $\gamma_n,\ n \notin \mathcal{Q}$, are mostly small values in this case, leading to insignificant approximation error of $\mathbf{G}$, reduced complexity, and negligible impact on ISTR performance.

Consequently, the above efficient implementation of ISTR is concluded in Algorithm 1, where T is the number of iterations. Since the complexity of signal detection is dominant in turbo receiver, by utilizing the above low-complexity computation strategies, the total computational complexity of the ISTR with the efficient implementation is given by

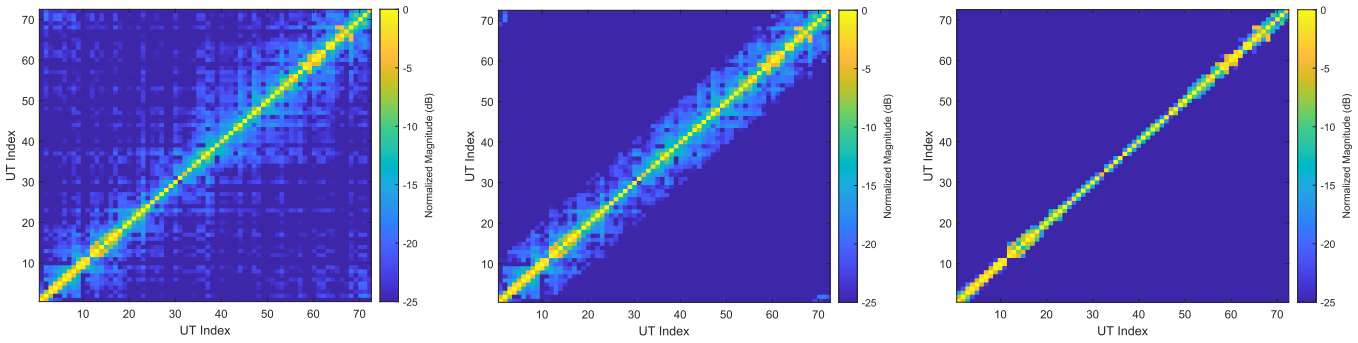
$$\begin{aligned} N_{\rm tot} &= N_{\mathbf{G}} + N_{\boldsymbol{\theta}} + T\left(N_{\mathbf{w}} + N_{\overline{\boldsymbol{\theta}}} + N_{\rm det}\right) \\ &= \widetilde{A}(G_{\rm nz} + U(Q+1))/2 + (S\log S)/2 + A + \widetilde{A}U \\ &\quad + T\left(G_{\rm nz} + U\left(\widetilde{N}^3/2 + 3\widetilde{N}^2/2 + 2\widetilde{N}\right)\right). \end{aligned} \tag{45}$$

B. Efficient Construction of IUS Using sCSI

According to Section III-C, the construction of the IUS for each UT relies on channel correlation $\eta_{i,j}$ between UTs, which is computed by $\mathbf{G}$ and updated as the iCSI, $\mathbf{H}$, changes. When UTs exhibit mobility or the ionosphere is relatively active, the time-variation of channels is significant, resulting in a high overhead for the updating of IUS. To tackle this issue, we turn to utilize the sCSI defined in BBCM, $\mathbf{R}_u$ or $\boldsymbol{\omega}_u$, for IUS construction. Motivated by the definition of channel correlation, we further define the statistical channel correlation between UT i and UT j as

$$\overline{\eta}_{i,j} \triangleq \frac{{\rm tr}\left\{\mathbf{R}_i^{\rm H}\mathbf{R}_j\right\}}{\|\mathbf{R}_i\|\|\mathbf{R}_j\|}. \tag{46}$$

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(a) (b) (c)

Fig. 4. Normalized magnitude of $\mathbf{G}$ in dB: (a) exactly computed $\mathbf{G}$ with $\varepsilon = 0$ ($Q = 128$), (b) approximately computed $\mathbf{G}$ with $\varepsilon = 0.01$ ($Q = 33$) and (c) approximately computed $\mathbf{G}$ with $\varepsilon = 0.1$ ($Q = 3$).

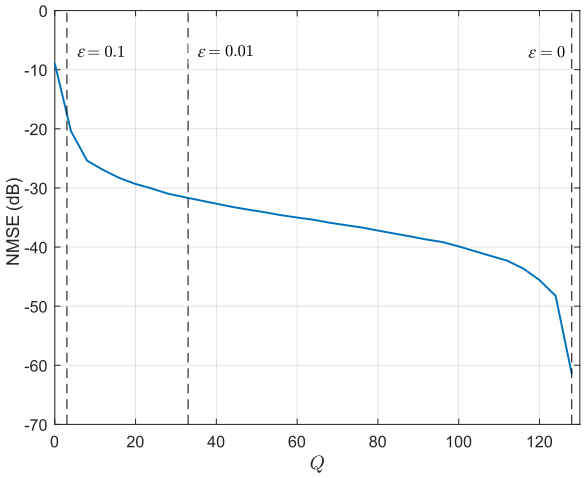


Fig. 5. NMSE in dB of computing $\mathbf{G}$ against the cardinality of the filtering set $\mathcal{Q}$.

Algorithm 1 Efficient Implementation of ISTR

Initialization $t = 1$, $L'_{\mathrm{a}}(q_{u,i}) = 0,\ \forall u \in \mathcal{Z}_U^+, i \in \mathcal{Z}_N^+$.
Compute $\mathbf{G} = \mathbf{D}^{\mathrm{H}}\mathbf{P}$ with $\mathbf{P}$ computed by (43).
Compute $\boldsymbol{\theta}$ by (39).
while $t \leq T$ **do**
  Compute $P(q_{u,i} = b)$, $\forall u \in \mathcal{Z}_U^+, \forall i \in \mathcal{Z}_N^+$, by (16) and update $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ by (17), respectively.
  Compute $\mathbf{w}_u,\ \forall u \in \mathcal{Z}_U^+$, by (38).
  Compute $\widehat{x}_u$, $\widehat{\epsilon}_u$, $\forall u \in \mathcal{Z}_U^+$, by (33).
  Compute $L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u)$, $\forall u \in \mathcal{Z}_U^+, \forall i \in \mathcal{Z}_N^+$, by (13) with $\widetilde{x}_u$ and $\widetilde{\epsilon}_u$ replaced by $\widehat{x}_u$ and $\widehat{\epsilon}_u$, respectively.
  If $t < T$, update $L'_{\mathrm{a}}(q_{u,i})$, $\forall u \in \mathcal{Z}_U^+, \forall i \in \mathcal{Z}_N^+$, after deinterleaving, SISO decoding and interleaving.
  Set $t \leftarrow t + 1$.
end
Output $\widehat{c}_{u,i}$, $\forall u \in \mathcal{Z}_U^+, \forall i \in \mathcal{Z}_N^+$, after deinterleaving and SISO decoding.

Similar to channel correlation $\eta_{i,j}$, for statistical channel correlation $\overline{\eta}_{i,j}$, we have $0 \leq \overline{\eta}_{i,j} \leq 1$, $\overline{\eta}_{i,j} = \overline{\eta}_{j,i}$, and $\overline{\eta}_{i,j} = 0$ when $\mathrm{tr}\left\{\mathbf{R}_i^{\mathrm{H}}\mathbf{R}_j\right\} = 0$, $\overline{\eta}_{i,j} = 1$ when $\mathbf{R}_i$ is a scaled version of $\mathbf{R}_j$.

To show the relationship between $\overline{\eta}_{i,j}$ and $\eta_{i,j}$, we rewrite the statistical channel correlation between UT i and UT j as $\overline{\eta}_{i,j} = \zeta_{i,j}/\sqrt{\zeta_{i,i}\zeta_{j,j}}$, where

$$\zeta_{i,j} = \mathrm{tr}\left\{\mathbf{R}_i^{\mathrm{H}}\mathbf{R}_j\right\} = \mathrm{tr}\left\{\mathbf{V}\boldsymbol{\Omega}_i\mathbf{V}^{\mathrm{H}}\mathbf{V}\boldsymbol{\Omega}_j\mathbf{V}^{\mathrm{H}}\right\} \overset{\text{(a)}}{=} \mathrm{tr}\left\{\mathbf{Q}\boldsymbol{\Omega}_i\mathbf{Q}\boldsymbol{\Omega}_j\right\} = \sum_{a\in\mathcal{A}_i}\sum_{a'\in\mathcal{A}_j}[\boldsymbol{\omega}_i]_a[\boldsymbol{\omega}_j]_{a'}\gamma^2_{|a-a'|}, \quad (47)$$

in which (a) follows from the fact that $\mathrm{tr}\{\mathbf{AB}\} = \mathrm{tr}\{\mathbf{BA}\}$. Similar to the expression of $g_{i,j}$ in (20), for UT i and UT j with considerable geographical separation, $\gamma^2_{|a-a'|}$ is close to zero, leading to $\zeta_{i,j} \approx 0$ and $\overline{\eta}_{i,j} \approx 0$. This, to a certain extent, confirms the rationality of using statistical channel correlation $\overline{\eta}_{i,j}$ as an effective metric for constructing the IUS. As a result, instead of (26), the IUS for UT u can be constructed by

$$\mathcal{N}_u \triangleq \left\{u' \in \mathcal{Z}_U^+ | \overline{\eta}_{u,u'} > \overline{\eta}_{\mathrm{th}}\right\} \text{ with } N_u = |\mathcal{N}_u|, \quad (48)$$

where $\overline{\eta}_{\mathrm{th}} < 1$ is a preset threshold.

Due to the slow-varying property of the sCSI, the IUS can be reused across different subcarriers and over a considerably long time period. The variation in sCSI is primarily driven by large-scale ionospheric dynamics. According to the Watterson model and classical HF channel measurements [53], the HF channel is typically considered wide-sense stationary (WSS) within a time window up to 10 minutes. Within this window, the IUS remains relatively stable. To address the potential mismatch caused by channel non-stationarity, the IUS update period should be set shorter than the time window of WSS. Compared to the iCSI-based method that requires IUS updates every coherence time (usually < 1 s), updating every few minutes reduces the computational overhead by several orders of magnitude.

V. Simulation Results

In this section, simulation results of the proposed ISTR for HF skywave massive MIMO are provided.

UTs are distributed within a sector spanning 150° and located approximately $2,000$ km away from the BS and isotropic antennas are equipped at both ends. Some physical channel parameters, such as path gain and delay for each path from each UT, are generated by the ray-tracing software

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TABLE II
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency f_c	16 MHz
Number of antennas at the BS M	256
Antenna spacing at the BS d	9 m
Number of UTs U	72
Modulation scheme	16-QAM
Channel code	3/4 LDPC code
Code length	2112

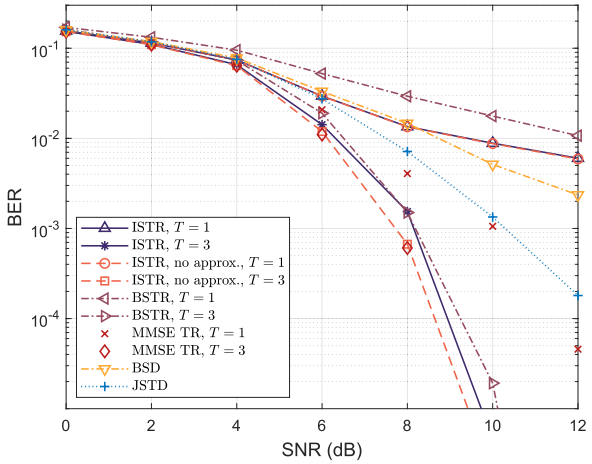


Fig. 6. BER performance of the ISTR with and without approximation, compared with the BSTR, MMSE TR, BSD and JSTD.

Proplab-Pro 3.1 [11]. The space domain channel at each subcarrier is generated accordingly. The beam domain iCSI and sCSI can be respectively obtained through the method proposed in [9] and [54] with fine factor $F = 2$. A row-column interleaver with length 528 is utilized for transmission. The maximum communication time spanned by the simulation is 2.1 s, during which the statistical CSI remains constant [53]. More detailed simulation parameters are presented in Table II.

We adopt the BER as the performance metric for receivers. Fig. 6 compares the BER performance of the proposed ISTR with and without approximation, where the BSTR [43], the MMSE turbo receiver (MMSE TR) [28], the beam structured detector (BSD) [12], and the joint Slepian transform based detector (JSTD) [11] are included for comparison and the SNR represents the received SNR. For ISTR, we use iCSI for interference selection and set $\eta_{\text{th}} = 0.03$ and $\varepsilon = 0.01$. Moreover, the ISTR with and without approximation refer to the ISTR using (43) and (38) for approximate computation and the ISTR using (41) and (32) for exact computation, respectively. From the figure, the BER performance of the ISTR with approximation approaches that of the ISTR without approximation. Especially when the number of iterations $T = 3$, both are slightly superior to that of the BSTR and approach that of the MMSE TR. Therefore, unless otherwise specified, the ISTR hereinafter refers to the ISTR with approximation. At the BER of 10^{-3}, the ISTR with $T = 3$ achieves approximately a 2 dB gain compared with the MMSE TR with $T = 1$ and the JSTD, and more than 4 dB gain compared with the BSD. The reason why the ISTR provides better performance than linear

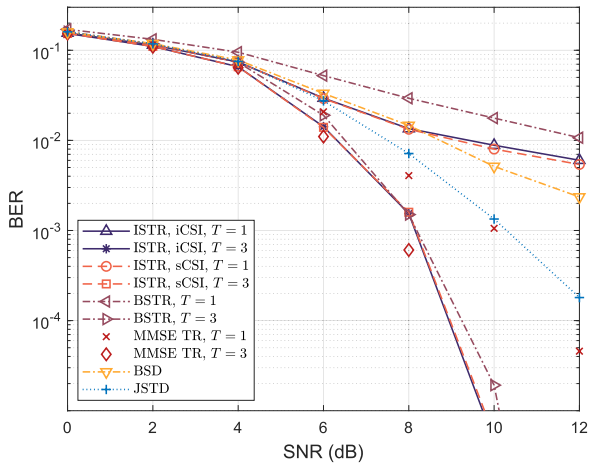


Fig. 7. BER performance of the ISTR with different IUS construction methods, compared with the BSTR, MMSE TR, BSD and JSTD.

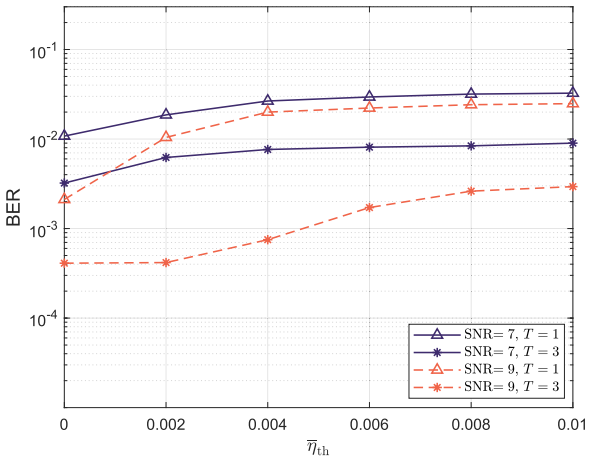
receivers (e.g., MMSE TR with $T = 1$, BSD, and JSTD) lies in its turbo iterations between the detector and decoder, which provide more accurate a prior information for the detector and updated soft input for the decoder alternatively.

Fig. 7 illustrates the BER performance of the ISTR with the construction of IUS using iCSI and sCSI, respectively, where $\varepsilon = 0.01$, $\eta_{\text{th}} = 0.03$, and $\overline{\eta}_{\text{th}} = 0.002$, in which case $\widetilde{N}$ keeps the same. The BER curves of the two nearly overlap with $T = 3$, slightly outperforming the BSTR and approaching that of the MMSE TR. This confirms the effectiveness of the construction of IUS using sCSI.

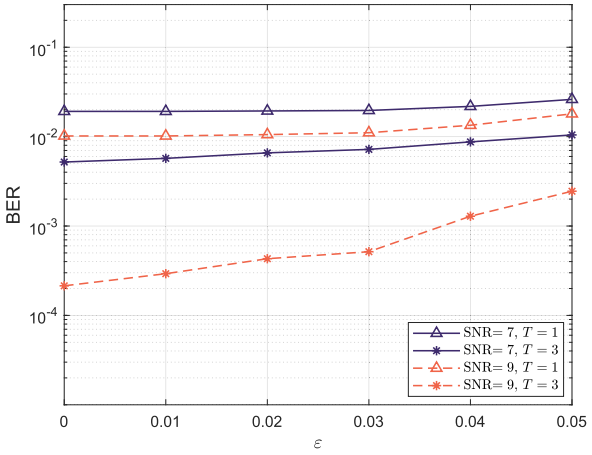
In Fig. 8, BER performance of the ISTR under various values of thresholds is presented, where sCSI is utilized for the construction of IUS. Fig. 8(a) plots the BER against different values of threshold $\overline{\eta}_{\text{th}}$, where $\varepsilon = 0.01$. The overall trend is that the BER performance deteriorates as $\overline{\eta}_{\text{th}}$ increases, which is consistent with the result in (29). Specially, for SNR = 9 dB with $T = 3$, the BER is lower than 10^{-3} when $\overline{\eta}_{\text{th}} \leq 0.004$, and Table III illustrates the values of the average size of IUS, $\widetilde{N}$, under different thresholds $\overline{\eta}_{\text{th}}$. Fig. 8(b) shows the BER against different values of threshold ε, where $\overline{\eta}_{\text{th}} = 0.002$. It can be seen that the BER performance degrades as ε grows, especially for $T = 3$. This can be attributed to the fact that a larger threshold ε renders the approximation of $\mathbf{G}$ more inaccurate, thereby degrading the BER performance. Moreover, Table IV presents the values of Q and $\rho_{\mathbf{G}}$ under different values of ε, in which $\rho_{\mathbf{G}} = G_{\text{nz}}/U^2 \times 100\%$ represents the percentage of non-zero elements in the approximately computed $\mathbf{G}$. From Table III and Table IV, larger values of thresholds $\overline{\eta}_{\text{th}}$ and ε respectively lead to smaller values of $\widetilde{N}$ and Q, yielding a lower computational complexity. To achieve a tradeoff between performance and complexity, $\overline{\eta}_{\text{th}} = 0.002$ and $\varepsilon = 0.01$ are suitable choices in this scenario, in which case 4 to 5 UTs are generally sufficient for the construction of IUS, demonstrating the effectiveness and robustness of the proposed ISTR.

To better examine the impact of different sizes of IUS on the BER performance of the ISTR, Fig. 9 plots the BER performance of the ISTR with different sizes of IUS under given thresholds $\overline{\eta}_{\text{th}} = 0.002$ and $\varepsilon = 0.01$. In Fig. 9, the

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(a)



(b)

Fig. 8. BER performance of the ISTR under various values of thresholds. (a) Threshold $\overline{\eta}_{\rm th}$ for the construction of IUS with $\varepsilon = 0.01$; (b) Threshold ε for approximate computation of $\mathbf{G}$ with $\overline{\eta}_{\rm th} = 0.002$.

TABLE III

VALUES OF $\widetilde{N}$ UNDER DIFFERENT THRESHOLDS $\overline{\eta}_{\rm th}$

$\overline{\eta}_{\rm th}$	0	0.002	0.004	0.006	0.008	0.01
$\widetilde{N}$	21.75	4.81	3.67	3.17	2.83	2.75

TABLE IV

VALUES OF Q, $G_{\rm nz}$ AND $\rho_{\mathbf{G}}$ UNDER DIFFERENT THRESHOLDS ε

ε	0	0.01	0.02	0.03	0.04	0.05
Q	128	33	16	11	8	6
$G_{\rm nz}$	5184	1566	870	678	558	462
$\rho_{\mathbf{G}}$	100.00%	30.21%	16.78%	13.08%	10.76%	8.91%

IUS $\mathcal{N}_u'$ for UT u is constructed by adding/removing UTs into/from the original IUS $\mathcal{N}_u$, where $\mathcal{N}_u$ is constructed by (48). From the figure, when 2 UTs are removed from each IUS, the BER performance is degraded, especially in high SNR region. This is because for high SNR, the noise power is small and the effect of inter-UT interference dominates, and removing UTs from the IUS will weaken the capability of interference suppression. Therefore, the size of IUS can be flexibly adjusted in accordance with the level of SNR. Moreover, when 2 UTs are added into each IUS, the BER performance is improved. Since the original IUS already includes the dominant interfering UTs, the performance gain of enlarging the IUS is marginal after several iterations.

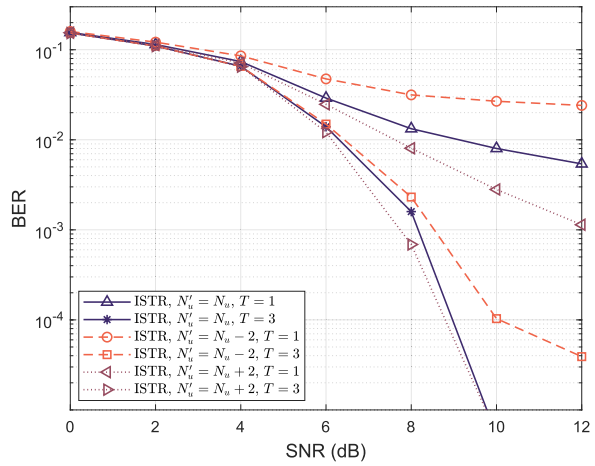


Fig. 9. BER performance of the ISTR with different sizes of IUS, where $\overline{\eta}_{\rm th} = 0.002$, $\varepsilon = 0.01$.

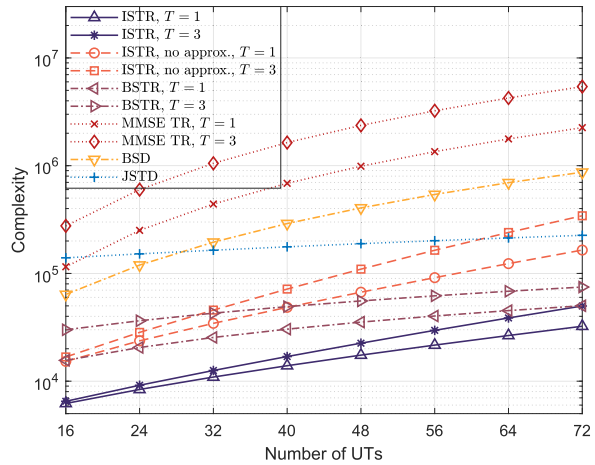


Fig. 10. Computational complexity of the ISTR, compared with the BSTR, MMSE TR, BSD and JSTD.

In Fig. 9, the computational complexities of the ISTR and the other receivers are compared versus the number of UTs, where $\overline{\eta}_{\rm th} = 0.002$ and $\varepsilon = 0.01$. For the ISTR without approximation, its complexity is below that of the MMSE TR and BSD and is lower than that of the JSTD in most cases, even with $T = 3$. Besides, compared with the ISTR without approximation and the BSTR, the ISTR with approximation exhibits a more pronounced complexity reduction and it is insensitive to the number of iterations.

With the aid of theoretical analysis and simulation results, we summarize the selection strategies for key system parameters to guide the practical deployment of the proposed ISTR:

- **Number of antennas:** In typical massive MIMO deployments, the antenna size is commonly on the order of hundreds. Therefore, we consider $M = 128$ or 256 as

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a typical and practically relevant choice, in which case the interference sparsity is remarkable and it provides a good balance between performance improvement and implementation cost.

- **IUS threshold:** Our simulation results indicate that selecting the threshold such that the IUS size for each UT is approximately 4-5 is sufficient to capture the dominant interference components in scenarios where UTs are geographically well separated. In this operating region, the main interfering UTs are already covered, and further enlarging the IUS yields only marginal performance improvements but increases computational complexity.
- **Number of turbo iterations:** While increasing the number of iterations can improve the system performance, the complexity grows approximately linearly with the iteration number, and the incremental performance gain tends to diminish after a few iterations. Based on our simulation results, 3 turbo iterations are sufficient for most cases considered, achieving a satisfactory performance while keeping the computational burden moderate.

VI. Conclusion

In this paper, we investigated receiver design for HF skywave massive MIMO communications. We first introduced the BBCM and showed the inherent beam domain channel sparsity and interference sparsity for HF skywave massive MIMO systems. Then, we discussed sufficient observation selection for signal detection by employing interference sparsity. As a result, the IUS was constructed and the ISTR was proposed with low-dimensional ISDs. The optimality of the ISTR was proved when a relaxed version of the condition for sufficient observation selection is satisfied. Next, we proposed an efficient implementation of the ISTR by adopting the properties of the beam matrix and various approximate computation strategies. Further, an efficient construction of IUS using sCSI was provided. Simulation results manifested the effectiveness of the proposed ISTR and some system design insights and implications are provided. Besides, the concept of interference sparsity can also be applied to other aspects of system design, such as user scheduling, downlink precoding, and channel estimation, which deserves investigation in future work.

Appendix A
Proof of Theorem 1

The proof will be organized as follows: Firstly, we will derive the expressions of $I(x_u,\mathbf{y})$ and $I(x_u,\boldsymbol{\theta}_u)$ using the definition of mutual information, respectively. Next, we will prove inequality (22). Finally, we will verify that (23) is a condition for the equality of (22).

Since $\mathbf{x}\sim\mathcal{CN}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ and $\mathbf{z}\sim\mathcal{CN}(\mathbf{0},\sigma_{\mathrm{z}}\mathbf{I})$, we have $x_u\sim\mathcal{CN}(\mu_u,\sigma_u)$ and $\mathbf{y}\sim\mathcal{CN}(\mathbf{H}\boldsymbol{\mu},\mathbf{C})$, where $\mathbf{C}=\mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^{\mathrm{H}}+\sigma_{\mathrm{z}}\mathbf{I}$. Then the entropies of x_u and $\mathbf{y}$ are [55]

$$H(x_u)=\frac{1}{2}\log(2\pi e\sigma_u), \tag{49a}$$

$$H(\mathbf{y})=\frac{1}{2}\log\left((2\pi e)^M\det(\mathbf{C})\right), \tag{49b}$$

respectively, where $\det(\mathbf{C})$ is the determinant of $\mathbf{C}$. We next define an augmented vector $\mathbf{t}_u\triangleq\left[x_u,\mathbf{y}^{\mathrm{T}}\right]^{\mathrm{T}}\in\mathbb{C}^{M+1}$. Then, the covariance matrix associated with $\mathbf{t}_u$ is

$$\mathrm{Cov}\{\mathbf{t}_u,\mathbf{t}_u\}=\begin{bmatrix}\sigma_u & \sigma_u\mathbf{h}_u^{\mathrm{H}}\\ \mathbf{h}_u\sigma_u & \mathbf{C}\end{bmatrix}. \tag{50}$$

Therefore, the joint entropy of x_u and $\mathbf{y}$ is given by

$$\begin{aligned}H(x_u,\mathbf{y})&=\frac{1}{2}\log\left((2\pi e)^{M+1}\det(\mathrm{Cov}\{\mathbf{t}_u,\mathbf{t}_u\})\right)\\&=\frac{1}{2}\log\left((2\pi e)^{M+1}\sigma_u\det\left(\mathbf{C}-\sigma_u\mathbf{h}_u\mathbf{h}_u^{\mathrm{H}}\right)\right).\end{aligned} \tag{51}$$

Thus, the mutual information between x_u and $\mathbf{y}$ is given by

$$\begin{aligned}I(x_u,\mathbf{y})&=H(x_u)+H(\mathbf{y})-H(x_u,\mathbf{y})\\&=-\frac{1}{2}\log\left(1-\sigma_u\mathbf{h}_u^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{h}_u\right).\end{aligned} \tag{52}$$

Moreover, it can be obtained that $\boldsymbol{\theta}_u\sim\mathcal{CN}(\mathbf{G}_u\boldsymbol{\mu},\mathbf{C}_u)$ with $\mathbf{C}_u=\mathbf{N}_u^{\mathrm{T}}\mathbf{H}^{\mathrm{H}}\mathbf{C}\mathbf{H}\mathbf{N}_u$, and

$$H(\boldsymbol{\theta}_u)=\frac{1}{2}\log\left((2\pi e)^{N_u}\det(\mathbf{C}_u)\right), \tag{53a}$$

$$H(x_u,\boldsymbol{\theta}_u)=\frac{1}{2}\log\left((2\pi e)^{N_u+1}\sigma_u\det\left(\mathbf{C}_u-\sigma_u\mathbf{G}_u\mathbf{e}_u\mathbf{e}_u^{\mathrm{T}}\mathbf{G}_u^{\mathrm{H}}\right)\right). \tag{53b}$$

Then, the mutual information between x_u and $\boldsymbol{\theta}_u$ can be similarly derived as

$$\begin{aligned}I(x_u,\boldsymbol{\theta}_u)&=H(x_u)+H(\boldsymbol{\theta}_u)-H(x_u,\boldsymbol{\theta}_u)\\&=-\frac{1}{2}\log\left(1-\sigma_u\mathbf{e}_u^{\mathrm{T}}\mathbf{G}_u^{\mathrm{H}}\mathbf{C}_u^{-1}\mathbf{G}_u\mathbf{e}_u\right).\end{aligned} \tag{54}$$

As a special case, when $\mathbf{N}_u=\mathbf{I}$,

$$I(x_u,\boldsymbol{\theta})=-\frac{1}{2}\log\left(1-\sigma_u\mathbf{e}_u^{\mathrm{T}}\mathbf{G}\left(\mathbf{H}^{\mathrm{H}}\mathbf{C}\mathbf{H}\right)^{-1}\mathbf{G}\mathbf{e}_u\right). \tag{55}$$

Since $\mathbf{G}\mathbf{e}_u=\mathbf{H}^{\mathrm{H}}\mathbf{h}_u$, then

$$\begin{aligned}\mathbf{e}_u^{\mathrm{T}}\mathbf{G}\left(\mathbf{H}^{\mathrm{H}}\mathbf{C}\mathbf{H}\right)^{-1}\mathbf{G}\mathbf{e}_u&=\mathbf{h}_u^{\mathrm{H}}\mathbf{H}(\boldsymbol{\Sigma}\mathbf{G}+\sigma_{\mathrm{z}}\mathbf{I})^{-1}\mathbf{e}_u\\&=\mathbf{h}_u^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{h}_u,\end{aligned} \tag{56}$$

which implies

$$I(x_u,\boldsymbol{\theta})=I(x_u,\mathbf{y}). \tag{57}$$

Additionally, according to data-processing inequality [51],

$$I(x_u,\boldsymbol{\theta}_u)\le I(x_u,\boldsymbol{\theta}). \tag{58}$$

Combining (57) and (58) yields (22).

We next verify that (23) is a condition for equality of (22). From (52) and (54), $I(x_u,\boldsymbol{\theta}_u)=I(x_u,\mathbf{y})$ is equivalent to $\rho_u=\varrho_u$ with $\rho_u=\mathbf{h}_u^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{h}_u$ and $\varrho_u=\mathbf{e}_u^{\mathrm{T}}\mathbf{G}_u^{\mathrm{H}}\mathbf{C}_u^{-1}\mathbf{G}_u\mathbf{e}_u$. When (23) holds, we have

$$\begin{aligned}\varrho_u&=\mathbf{e}_u^{\mathrm{T}}\mathbf{G}_u^{\mathrm{H}}\left(\mathbf{N}_u^{\mathrm{T}}\left(\mathbf{G}\boldsymbol{\Sigma}\mathbf{G}+\sigma_{\mathrm{z}}\mathbf{G}\right)\mathbf{N}_u\right)^{-1}\mathbf{G}_u\mathbf{e}_u\\&=\mathbf{e}_u^{\mathrm{T}}\mathbf{G}_u^{\mathrm{H}}\left(\mathbf{G}_u'\boldsymbol{\Sigma}_u\mathbf{G}_u'+\sigma_{\mathrm{z}}\mathbf{G}_u'\right)^{-1}\mathbf{G}_u\mathbf{e}_u\\&=\mathbf{e}_u^{\mathrm{T}}\mathbf{G}_u^{\mathrm{H}}\left(\boldsymbol{\Sigma}_u\mathbf{G}_u'+\sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u,\end{aligned} \tag{59}$$

where $\mathbf{G}_u=\mathbf{N}_u^{\mathrm{T}}\mathbf{G}$, $\mathbf{G}_u'=\mathbf{N}_u^{\mathrm{T}}\mathbf{G}\mathbf{N}_u$ and $\boldsymbol{\Sigma}_u=\mathbf{N}_u^{\mathrm{T}}\boldsymbol{\Sigma}\mathbf{N}_u$. It can be checked that

$$\mathbf{N}_u\left(\boldsymbol{\Sigma}_u\mathbf{G}_u'+\sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}=\left(\boldsymbol{\Sigma}\mathbf{G}+\sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}\mathbf{N}_u. \tag{60}$$

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Therefore,

$$\varrho_u = \mathbf{e}_u^{\mathrm{T}}\mathbf{G}\left(\boldsymbol{\Sigma}\mathbf{G}+\sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}\mathbf{N}_u\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u = \mathbf{h}_u^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{h}_u = \rho_u. \tag{61}$$

Consequently,

$$I(x_u,\boldsymbol{\theta}_u) = -\frac{1}{2}\log\left(1-\sigma_u\rho_u\right) = I(x_u,\mathbf{y}). \tag{62}$$

This completes the proof.

Appendix B
Proof of Theorem 2

From III-A, $L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$ is calculated by MMSE detector. And the difference between $L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u)$ and $L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$ comes from the detection output of x_u. Therefore, we turn to prove that under the given condition, $\widehat{x}_u = \widetilde{x}_u$ and $\widehat{\epsilon}_u = \widetilde{\epsilon}_u$.

When $u \in \mathcal{N}_u$ and $\forall l \notin \mathcal{N}_u$, $g_{k,l} = 0$, $\forall k \in \mathcal{N}_u$ or $\sigma_l = 0$, we have

$$\sigma_l g_{k,l} = 0, \quad \forall k \in \mathcal{N}_u, l \notin \mathcal{N}_u, \tag{63}$$

then the following relations hold:

$$\boldsymbol{\Sigma}\mathbf{G}\mathbf{N}_u = \mathbf{N}_u\boldsymbol{\Sigma}\mathbf{N}_u^{\mathrm{T}}\mathbf{G}\mathbf{N}_u = \mathbf{N}_u\boldsymbol{\Sigma}_u\mathbf{G}_u', \tag{64a}$$

$$\mathbf{G}_u\boldsymbol{\Sigma}\mathbf{G}_u^{\mathrm{H}} = \mathbf{G}_u\boldsymbol{\Sigma}\mathbf{G}\mathbf{N}_u = \mathbf{G}_u'\boldsymbol{\Sigma}_u\mathbf{G}_u'. \tag{64b}$$

Followed by (64), $\mathbf{r}_u$ and $\mathbf{w}_u$ can be respectively written as

$$\begin{aligned}\mathbf{r}_u &= \mathbf{H}(\boldsymbol{\Sigma}\mathbf{G}+\sigma_{\mathrm{z}}\mathbf{I})^{-1}\mathbf{N}_u\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u\sigma_u \\ &= \mathbf{H}\mathbf{N}_u\left(\boldsymbol{\Sigma}_u\mathbf{G}_u'+\sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u\sigma_u, \end{aligned}\tag{65a}$$

$$\begin{aligned}\mathbf{w}_u &= \left(\mathbf{N}_u^{\mathrm{T}}\left(\mathbf{G}\boldsymbol{\Sigma}\mathbf{G}+\sigma_{\mathrm{z}}\mathbf{G}\right)\mathbf{N}_u\right)^{-1}\mathbf{G}_u\mathbf{e}_u\sigma_u \\ &= \left(\boldsymbol{\Sigma}_u\mathbf{G}_u'+\sigma_{\mathrm{z}}\mathbf{I}\right)^{-1}\mathbf{N}_u^{\mathrm{T}}\mathbf{e}_u\sigma_u. \end{aligned}\tag{65b}$$

Therefore, $\mathbf{r}_u = \mathbf{H}\mathbf{N}_u\mathbf{w}_u$. Consequently,

$$\widehat{x}_u = \mathbf{w}_u^{\mathrm{H}}\mathbf{N}_u^{\mathrm{T}}\mathbf{H}^{\mathrm{H}}(\mathbf{y}-\mathbf{H}\boldsymbol{\mu})+\mu_u = \mathbf{r}_u^{\mathrm{H}}(\mathbf{y}-\mathbf{H}\boldsymbol{\mu})+\mu_u = \widetilde{x}_u, \tag{66a}$$

$$\widehat{\epsilon}_u = \left(1-\mathbf{w}_u^{\mathrm{H}}\mathbf{N}_u^{\mathrm{T}}\mathbf{H}^{\mathrm{H}}\mathbf{h}_u\right)\sigma_u = \left(1-\mathbf{r}_u^{\mathrm{H}}\mathbf{h}_u\right)\sigma_u = \widetilde{\epsilon}_u. \tag{66b}$$

As a result, $L_{\mathrm{e}}(q_{u,i}|\boldsymbol{\theta}_u) = L_{\mathrm{e}}(q_{u,i}|\mathbf{y})$. This completes the proof.

Appendix C
Proof of Lemma 1

Exploiting (35), $\mathbf{Q}$ can be rewritten as

$$\mathbf{Q} = -\frac{1}{M}\mathbf{I}_{A,S}\overline{\boldsymbol{\Pi}}_S^{(A-1)/2}\mathbf{T}\overline{\boldsymbol{\Pi}}_S^{S-(A-1)/2}\mathbf{I}_{S,A}, \tag{67}$$

with $\mathbf{T} = \boldsymbol{\Lambda}^*\mathbf{F}_S^{\mathrm{H}}\mathbf{I}_{S,M}\mathbf{I}_{M,S}\mathbf{F}_S\boldsymbol{\Lambda}$. Since $\mathbf{I}_{S,M}\mathbf{I}_{M,S}$ is a diagonal matrix, then $\mathbf{F}_S^{\mathrm{H}}\mathbf{I}_{S,M}\mathbf{I}_{M,S}\mathbf{F}_S = \sum_{n=0}^{S-1}\beta_n\boldsymbol{\Pi}_S^n$ is a circulant matrix with

$$\beta_n = \mathbf{f}_{S,n+1}^{\mathrm{H}}\mathbf{I}_{S,M}\mathbf{1} = M\exp\{\bar{\jmath}\pi n(M-1)/S\}\gamma_n. \tag{68}$$

Noting that $\boldsymbol{\Lambda} = \mathrm{diag}\{\boldsymbol{\lambda}\}$ with $\boldsymbol{\lambda} = \mathbf{I}_{S,2S}\mathbf{f}_{2S,M}^*$, then

$$\begin{aligned}\mathbf{T} &= \mathrm{diag}\{\boldsymbol{\lambda}^*\}\sum_{n=0}^{S-1}\beta_n\boldsymbol{\Pi}_S^n\mathrm{diag}\{\boldsymbol{\lambda}\} \\ &= \left(\boldsymbol{\lambda}^*\boldsymbol{\lambda}^{\mathrm{T}}\right)\odot\sum_{n=0}^{S-1}\beta_n\boldsymbol{\Pi}_S^n. \end{aligned}\tag{69}$$

It can be easily derived that $\boldsymbol{\lambda}^*\boldsymbol{\lambda}^{\mathrm{T}} = \sum_{n=0}^{S-1}\exp\{-\bar{\jmath}\pi n(M-1)/S\}\overline{\boldsymbol{\Pi}}_S^n$. Combining the fact that $\boldsymbol{\Pi}_S^m\odot\overline{\boldsymbol{\Pi}}_S^n = \delta[\langle m-n\rangle_S]\overline{\boldsymbol{\Pi}}_S^n$, we have $\mathbf{T} = M\sum_{n=0}^{S-1}\gamma_n\overline{\boldsymbol{\Pi}}_S^n$, then we arrive at

$$\begin{aligned}\mathbf{Q} &= -\frac{1}{M}\mathbf{I}_{A,S}\overline{\boldsymbol{\Pi}}_S^{(A-1)/2}\left(M\sum_{n=0}^{S-1}\gamma_n\overline{\boldsymbol{\Pi}}_S^n\right)\overline{\boldsymbol{\Pi}}_S^{S-(A-1)/2}\mathbf{I}_{S,A} \\ &\overset{(a)}{=} \mathbf{I}_{A,S}\sum_{n=0}^{S-1}\gamma_n\overline{\boldsymbol{\Pi}}_S^n\mathbf{I}_{S,A}, \end{aligned}\tag{70}$$

where (a) follows from $\overline{\boldsymbol{\Pi}}_S^{(A-1)/2}\overline{\boldsymbol{\Pi}}_S^n\overline{\boldsymbol{\Pi}}_S^{S-(A-1)/2} = -\overline{\boldsymbol{\Pi}}_S^n$. Due to the property $\gamma_n = -\gamma_{S-n}$, $\langle n\rangle_S \neq 0$, $\mathbf{Q}$ can be rewritten as (40). This completes the proof.

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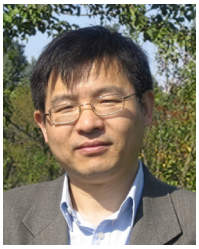
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